

Smart Congestion Relief

Comprehensive Evaluation of Traffic Congestion Costs and Congestion Reduction Strategies

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Abstract

How traffic congestion is evaluated can significantly affect transport planning decisions. This report investigates the best methods for measuring congestion costs and evaluating potential congestion reduction strategies. Key factors include analysis scope, baseline speeds, travel time valuation, accident and emission impact analysis, induced travel analysis, and consideration of co-benefits. It discusses how these factors influence planning decisions and describes the practices recommended by experts. It applies these methods to evaluate various congestion reduction strategies, including roadway expansion, improvement of space efficient modes, pricing reforms, Smart Growth policies and demand management programs.

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Table of Contents

Executive Summary	3
Introduction	8
Context: Changing Travel Demands and a New Planning Paradigm	11
Quantifying and Monetizing Congestion Costs	15
Analysis Scope	15
Baseline Speeds	17
Data Collection	19
Travel Time Values	19
Perspective.....	19
Fuel Consumption and Emission Impacts.....	20
Safety Impacts.....	20
New Technologies.....	21
Generated Traffic and Induced Travel	21
Congestion Cost Evaluation	22
Guidelines for Comprehensive and Multimodal Congestion Evaluation	27
Accessibility Analysis	27
Comprehensive Impact Analysis.....	28
Economic Efficiency and Consumer Surplus	30
Social Equity.....	32
Comprehensive Evaluation Summary.....	33
Alternatives to Roadway Level-Of-Service	34
Multimodal Level-of-Service (LOS) and Quality of Service (QOS)	34
Trip Generation, Vehicle Travel and Fuel Consumption Models	34
Multimodal Accessibility Modeling.....	35
Measuring Efficiency	37
Optimizing Urban Accessibility	40
Current Congestion Evaluation.....	41
Economic Productivity Impacts.....	43
Evaluating Potential Congestion Reduction Strategies	47
Roadway Capacity Expansion	47
Improving and Favoring Space-Efficient Modes	49
Transport Pricing Reforms	57
Improving Traveler Information	59
Smart Growth Development Policies	60
TDM Programs	61
Summary of Congestion Evaluation Strategies	62
What Does Modeling Indicate?.....	63
Optimal Congestion Solutions	64
Examples	65
Conclusions	67
References.....	69

Executive Summary

Traffic congestion refers to the incremental delay caused by interactions among vehicles on a roadway as traffic volumes approach a roadway's capacity. This report investigates the best methods for measuring these costs and evaluating potential congestion reduction strategies.

How congestion is measured significantly affects its estimated magnitude. Some indicators, such as roadway Level-Of-Service (LOS), and the Travel Time Index (TTI) measure *congestion intensity*; the differences in traffic speeds between peak and off-peak periods. Such information is useful for making short-term decisions, such as how to travel across town during rush hour, but is unsuited for strategic transportation and land use planning decisions that affect both *congestion intensity* and *exposure* (the amount that people must drive under congested conditions). Comprehensive indicators measure *congestion costs*, which takes both of these factors into account.

For example, the TTI rates congestion worse in New York than in Houston, although per capita congestion costs are higher in Houston than New York, as illustrated below. In compact and multimodal cities, congestion is more intense but residents suffer less overall because they have better travel options and drive less during peak periods.

Figure ES-1 Congestion Indicators Compared

Congestion Intensity (Travel Time Index)	Congestion Costs (Delay Hours Per Commuter)
1. Los Angeles-Long Beach-Santa Ana CA (1.37)	1. Los Angeles-Long Beach-Santa Ana CA (44.9)
2. New York-Newark NY-NJ-CT (1.33)	2. Washington DC-VA-MD (44.3)
3. Washington DC-VA-MD (1.32)	3. Houston TX (41.0)
4. Boston MA-NH-RI (1.28)	4. Atlanta GA (39.4)
5. Houston TX (1.26)	5. San Francisco-Oakland CA (37.7)
6. Philadelphia PA-NJ-DE-MD (1.26)	6. Dallas-Fort Worth-Arlington TX (36.6)
7. Seattle WA (1.26)	7. Miami FL (36.5)
8. Dallas-Fort Worth-Arlington TX (1.26)	8. Boston MA-NH-RI (36.3)
9. Chicago IL-IN (1.25)	9. Chicago IL-IN (36.2)
10. Miami FL (1.25)	10. Philadelphia PA-NJ-DE-MD (35.4)
11. Atlanta GA (1.24)	11. Detroit MI (33.6)
12. San Francisco-Oakland CA (1.22)	12. Seattle WA (33.4)
13. Detroit MI (1.18)	13. New York-Newark NY-NJ-CT (29.7)

More compact urban regions (blue) tend to have more intense congestion but lower congestion costs than sprawled, auto-oriented regions (red). Rankings change depending on which indicator is used.

Described more generally, urban transportation planning is affected by whether the analysis measures *mobility* (travel speed) or *accessibility* (time and money required to reach desired services and activities). Planning decisions often involve trade-offs between them. For example, roadway expansions may reduce vehicle traffic delay but increase walking and bicycling delay. Wider roads also tend to stimulate more dispersed development, which increases travel distances. Ignoring these impacts exaggerates roadway expansion benefits and undervalues other congestion reduction strategies that improve transport options and land use accessibility, not just mobility.

Traffic congestion tends to maintain self-limiting equilibrium: it increases to the point that delays limit further peak-period vehicle travel. Unless roads are managed to favor space-efficient modes, rational travellers will shift from space-efficient modes, such as buses and rideshare vehicles, to driving alone, increasing total congestion delay (Curiel, et al. 2021). Congestion is to motorists what tides are to sailors, a factor that must be considered, for example, when deciding when, where and how to travel.

Proposals for infill development are often opposed on the grounds that they will increase congestion, but that is not necessarily true. Adding more people to an area increase *trips*, but with good planning this is offset by shorter trip distances and reduced automobile mode shares. Although congestion may become intense, per capita delays tend to decline, particularly if infill development is accompanied with improvements to space-efficient modes, TDM incentives such as efficient parking pricing and commute trip reduction programs, and with more mixed development so most common services are available within each neighborhood.

This report examines various methodological factors that affect congestion evaluation. These include the selection of *baseline speeds* (the traffic speeds below which delay costs are calculated), travel time unit costs (dollars per hour assigned to congestion delay), assumptions about how speed affects vehicle fuel consumption and emission rates, consideration of generated and induced vehicle travel, and the scope of indirect impacts considered when evaluating potential congestion reduction strategies. Experts recommend the following for more accurate and comprehensive congestion evaluation:

- Evaluate transport system performance based on overall *accessibility* (people's overall ability to reach desired services and activities) rather than just *mobility* (travel speed).
- Measure congestion *costs* rather than *intensity*. Intensity indicators, such as roadway LOS and the TTI, do not account for *exposure* (the amount that residents must drive during peak periods) and therefore total congestion costs.
- Measure delays to all travelers, not just to motorists. Account for pedestrian and cycling delays caused by wider roads and increased vehicle traffic (called the *barrier effect*), and the congestion avoided when travelers shift to public transit.
- Report the congestion costs travellers *impose* rather than just the costs they *bear*, when calculating efficient road prices or comparing the congestion costs of different modes.
- Use efficiency-optimizing rather than freeflow baseline speeds. Moderate traffic speeds (typically 40-50 miles per hour) maximize roadway throughput and fuel economy, so moderate congestion (LOS C) is usually most efficient overall. Freeflow speeds often exceed legal speed limits so a significant portion of the estimated congestion costs consists of traffic speed compliance (reducing speed to legal limits).
- Use travel time values that reflect users' actual willingness-to-pay for incremental speed gains. This is typically 20-40% of average wages for personal travel, and total wage, benefits, equipment and product time costs for commercial travel.

- Recognize variations in travel time values, and therefore, the efficiency gains provided by policies that favor higher value trips over lower-value trips. This tends to justify more freight and high-occupant vehicle priority strategies, and efficient road pricing.
- Recognize that congestion tends to maintain equilibrium: it increases to the point that delays limit further peak-period vehicle travel.
- Account for *generated* and *induced* vehicle travel (additional vehicle travel resulting from reduced congestion) when evaluating roadway expansions. Generated traffic tends to reduce long-term congestion reduction benefits, and induced travel tends to increase external costs including downstream congestion, crash risk and pollution emissions.
- Account for the increased crash costs that may result if congestion reductions increase traffic speeds or total vehicle travel.
- Account for co-benefits when evaluating potential congestion reduction strategies. Strategies that improve non-auto modes or efficiently price travel also tend to reduce parking costs, provide consumer savings, improve accessibility for non-drivers, increase safety and health, reduce pollution emissions, and support strategic land use objectives.
- Account for data biases. For example, Inrix and TomTom indices oversample the most congested roadways and so exaggerate average motorists' congestion costs.

Table ES-2 compares five types of congestion reduction strategies according to their congestion impacts, other costs and benefits, and degree they are considered in current planning.

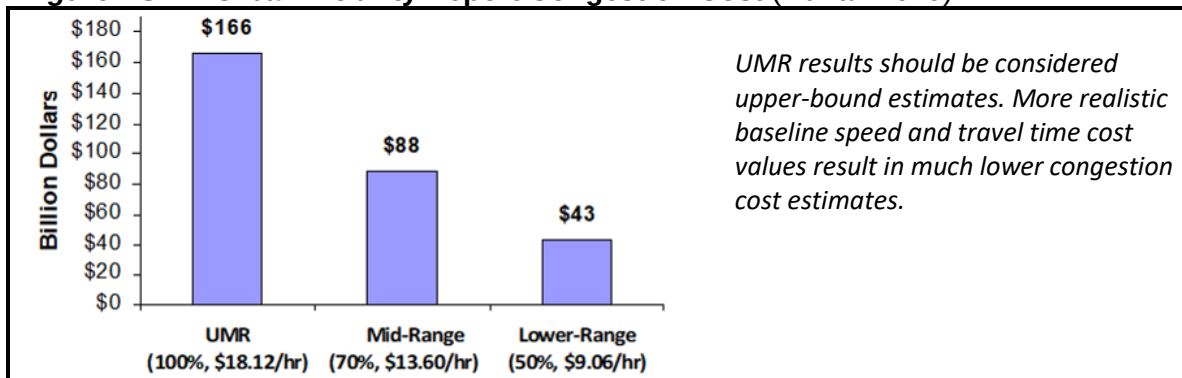
Table ES-2 Congestion Reduction Strategies

	Roadway Expansion	Improve Alt. Modes	Pricing Reforms	Smart Growth	TDM Programs
Congestion impacts	Reduces congestion in the short-run, but this declines over time due to generated traffic.	Reduces but does not eliminate congestion.	Can significantly reduce congestion.	May increase local congestion intensity but reduces per capita congestion costs.	Can reduce congestion delays and the costs to users of those delays.
Other costs and benefits	High costs. Minimal co-benefits. Tends to increase indirect costs by inducing vehicle travel.	Medium to high costs. Numerous co-benefits.	Low to high costs. User costs are offset by revenue generation. Many co-benefits.	Low to high costs. Numerous co-benefits.	Generally low to moderate implementation costs. Numerous co-benefits.
Consideration in current planning	Commonly considered and funded.	Sometimes considered, particularly in large cities.	Sometimes considered but seldom implemented.	Not generally considered a congestion reduction strategy.	Sometimes considered, particularly in large cities.

Different congestion reduction strategies have different types of impacts and benefits.

This study finds that many commonly-used congestion cost estimates are biased. For example, the *Urban Mobility Report* uses freeflow baseline speeds, excessive travel time values, and optimistic fuel saving and emission reduction estimates, and so represents an upper-bound value. More realistic assumptions result in significantly lower cost estimates. These biases tend to exaggerate congestion costs and roadway expansion benefits.

Figure ES-1 Urban Mobility Report Congestion Cost (Litman 2019)



Many congestion reduction strategies have been tried and failed, including roadway expansions, access management, high occupancy vehicle lanes, development restrictions, and commute trip reduction programs (Wachs, Chesney and Hwang 2020). Many jurisdictions are implementing innovative congestion reduction strategies, but few have implemented the optimal set, considering all impacts. To be effective a congestion reduction program must be multimodal: it must improve space-efficient modes *and* give traveller sufficient incentive to use those modes for a portion of peak-period trips.

An optimal congestion reduction program involves the following steps:

1. Improve space-efficient transport options, including walking, cycling, public transit, ridesharing, carsharing and telecommuting, so travellers can choose the most suitable for each trip. Target improvements to congested corridors. For example, improve transit services on congested corridors and implement TDM programs at major urban centers.
2. Manage roadways to favor space-efficient modes. These include transit-priority control systems as well as bus and High Occupant Vehicle (HOV) lanes on major roadways.
3. Implement support programs such as commute trip reduction and mobility management marketing programs wherever appropriate.
4. Apply decongestion pricing (road tolls that are higher during congested periods), with prices that maintain optimal traffic volumes (LOS C). If possible, apply prices system-wide, otherwise apply them on congested corridors such as urban highways and city centers. Revenues can help improve space-efficient modes or reduce other taxes.
5. Regardless of whether or not decongestion pricing is applied, implement pricing reforms such as revenue generating tolls, parking pricing, fuel price increases, and distance-based insurance and registration fees. These reforms are justified for efficiency and fairness.
6. Only consider urban roadway expansions if, after all of the previous strategies are fully implemented, all project costs can be recovered by user fees, which tests users' willingness-to-pay for the additional capacity. For example, if a roadway expansion would have \$5 million annualized costs, it should be implemented only if peak-period tolls on that road will generate that much revenue.

This is a timely issue. Current trends are increasing the importance of more comprehensive congestion analysis. Many jurisdictions are shifting from *mobility-* to *accessibility-oriented* transport planning, which recognizes that vehicle traffic speeds are just one of many factors that affect accessibility, and so are using more comprehensive indicators of transport system performance. Some are shifting from *LOS* (which assumes that the planning goal is maximize vehicle traffic speeds, so traffic congestion is the primary transportation problem) to *VMT* (which assumes that the planning goal is to reduce total vehicle miles of travel by improving alternative modes and creating more compact, multimodal communities). It is important that decision makers and the general public understand these issues when choosing solutions to congestion problems.

Introduction

Traffic congestion refers to travel delay caused by interactions between vehicles on a roadway, particularly as traffic volumes approach a roadway's capacity. There are many possible ways to measure congestion costs and evaluate potential solutions; a congestion reduction strategy may seem effective and desirable if evaluated one way, but ineffective and harmful if evaluated another. It is important that people involved in such decisions understand these issues.

For example, compact, multimodal cities such as New York, Boston and Philadelphia tend to have more *intense* congestion (greater peak-period speed reductions), but lower *congestion costs* (fewer annual delay-hours per capita). This results from the compact cities' lower auto mode shares and shorter trip lengths, which reduces congestion *exposure* (the amount residents must drive during peak periods). More dispersed, automobile-oriented cities such as Houston, Atlanta and Detroit tend to have less intense congestion but greater congestion costs. As a result, compact cities rank worse if evaluated by congestion intensity indicators, such as the Travel Time Index (TTI), but better if evaluated by congestion costs, as shown in tables 1 and 2.

Table 1 Congestion Indicators Compared (TTI 2015; ACS 2009)

Urban Region	Travel Time Index	Rank	Delay Per Auto Commuter (hrs.)	Auto Commute Mode Share	Delay Per Commuter (hrs.)	Rank
Los Angeles, CA	1.43	1	80	73%	58	1
San Francisco, CA	1.41	2	78	62%	48	3
Seattle, WA	1.38	3	63	70%	44	6
Washington DC	1.34	4	82	66%	54	2
New York, NY-NJ-CT	1.34	5	74	50%	37	13
Houston, TX	1.33	6	61	79%	48	4
Chicago, IL-IN	1.31	7	61	71%	43	8
Boston, MA-NH-RI	1.29	8	64	69%	44	5
Miami, FL	1.29	9	52	78%	41	10
Dallas, TX	1.27	10	53	81%	43	9
Phoenix, AZ	1.27	11	51	76%	39	12
Detroit, MI	1.24	12	52	84%	44	7
Atlanta, GA	1.24	13	52	77%	40	11
Philadelphia, PA-NJ-DE	1.24	14	48	74%	36	14
San Diego, CA	1.24	15	42	76%	32	15

Based on the Travel Time Index, New York ranks worse than Houston, but based on annual congestion delay hours per commuter, Houston ranks much worse than New York.

Table 2 City Rankings Change Depending on Indicators (TTI 2013)

Congestion Intensity (Travel Time Index)	Congestion Costs (Delay Hours Per Commuter)
14. Los Angeles-Long Beach-Santa Ana CA (1.37)	14. Los Angeles-Long Beach-Santa Ana CA (44.9)
15. New York-Newark NY-NJ-CT (1.33)	15. Washington DC-VA-MD (44.3)
16. Washington DC-VA-MD (1.32)	16. Houston TX (41.0)
17. Boston MA-NH-RI (1.28)	17. Atlanta GA (39.4)
18. Houston TX (1.26)	18. San Francisco-Oakland CA (37.7)
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21. Dallas-Fort Worth-Arlington TX (1.26)	21. Boston MA-NH-RI (36.3)
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26. Detroit MI (1.18)	26. New York-Newark NY-NJ-CT (29.7)
27. San Diego CA (1.18)	27. San Diego CA (28.0)
28. Phoenix-Mesa AZ (1.18)	28. Phoenix-Mesa AZ (26.7)

More compact urban regions (blue) tend to have more intense congestion but lower congestion costs than sprawled, auto-oriented regions (red). Rankings change depending on which indicator is used.

Congestion intensity indicators are useful for making short-term decisions, such as how best to travel across town during rush hour, but are unsuitable for strategic planning decisions that affect congestion *exposure*, the amount that travelers must drive under urban-peak conditions. To successfully evaluate decisions that affect the quality of travel options or development patterns requires the use of more comprehensive and multimodal analysis.

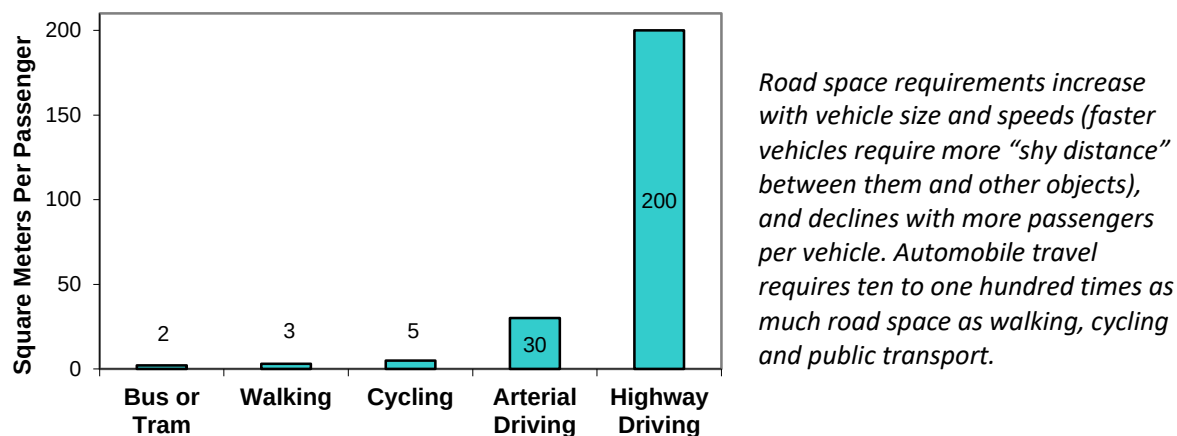
Described differently, intensity indicators reflect *mobility* (travel speed), while cost indicators reflect *accessibility* (people's overall ability to reach desired services and activities). Since accessibility is the ultimate goal of most transport activity and planning decisions often involve trade-offs between different accessibility factors, congestion cost indicators are most appropriate for identifying optimal transport system improvements.

Consider two examples. Assume that converting a general traffic lane into a bus lane reduces 10 minutes of delay for 20 buses carrying 1,000 passengers, but adds 5 minutes of delay for 800 cars carrying 900 passengers. If evaluated using congestion intensity indicators, the bus lanes are considered to reduce transport system performance because delay per vehicle increases. However, if evaluated based on congestion costs, it is considered to improve performance, since delay per passenger declines.

Similarly, conventional traffic impact studies often indicate that infill development reduces transport system performance, measured using roadway LOS. As a result, such projects are discouraged and burdened with special impact fees that are not imposed on the urban-fringe. This favors lower-density, automobile-dependent development. A more comprehensive evaluation, which accounts for the improved accessibility of infill development, and the resulting reductions in vehicle trip generation and trip distance rates, justifies more support and lower impact fees for such development.

Conventional evaluation often only measures the congestion costs that travelers *bear*, but some analyses, such as calculating mode shift benefits or optimal congestion reduction tolls, also require calculating the congestion costs that travelers *impose*. Different modes of travel have varying road space requirements resulting in their associated congestion costs (Figure 1).

Figure 1 Typical Road Space Requirements for Various Modes



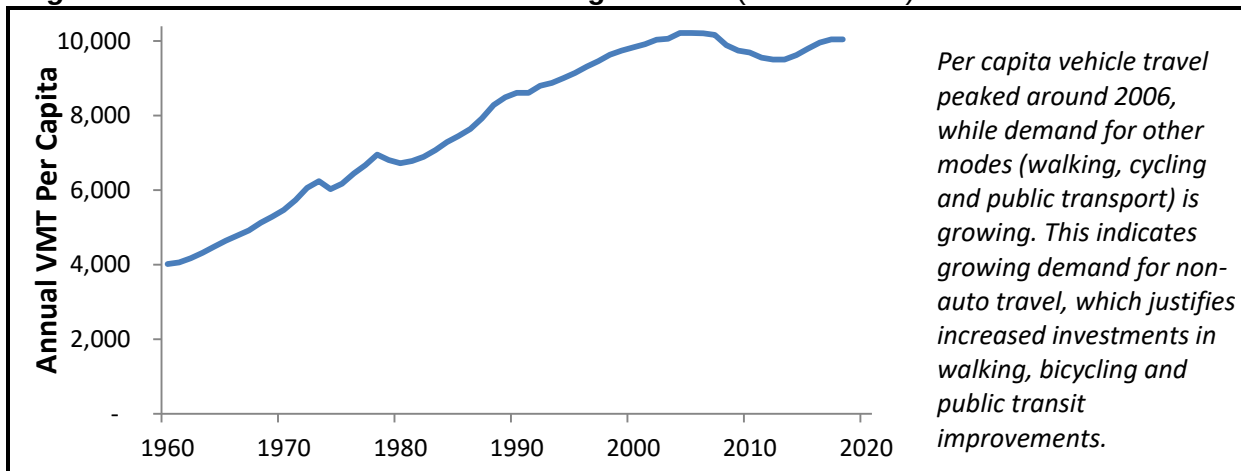
This is a timely issue. Current trends are increasing the importance of more comprehensive congestion analysis. Many jurisdictions are shifting from *mobility-* to *accessibility-oriented* transport planning, which recognizes that vehicle traffic speeds are just one of many factors that affect accessibility, and so are using more comprehensive indicators of transport system performance (Litman 2013). Some are shifting from *LOS* (which assumes that the planning goal is maximize vehicle traffic speeds, so traffic congestion is the primary transportation problem) to *VMT* (which assumes that the planning goal is to reduce total vehicle miles of travel by improving alternative modes and creating more compact, multimodal communities) (F&P 2019). It is important that decision makers and the general public understand these issues when choosing solutions to congestion problems.

In recent years, experts have developed more accurate and comprehensive congestion evaluation methods, but outdated practices are still widely used, and decision makers are often unaware of the biases in their results (Brasuell 2022; Metz 2021; Nguyen-Phuoc, et al. 2020). This report investigates these issues. It discusses various ways to define and measure congestion impacts, and the implications of different perspectives and methods. It describes best practices for measuring congestion costs and evaluating potential congestion reduction options, recommends ways to identify the most beneficial set of congestion reduction strategies, and describes examples of successful congestion reduction programs.

Context: Changing Travel Demands and a New Planning Paradigm

Transportation planning must respond to changing demands and community goals. Motor vehicle travel demand grew steadily during the twentieth century, so it made sense to invest in roadways. During that period there was little risk of overbuilding since any additional road capacity would soon fill. Vehicle travel is now peaking in most developed countries, and current demographic and economic trends (aging population, rising fuel prices, urbanization, increasing health and environmental concerns, and changing consumer preferences) are increasing demand for other transport options (Litman 2006; OECD 2012).

Figure 2 U.S. Annual Vehicles Mileage Trends (FHWA 2019)



Transport planning is experiencing a *paradigm shift*, a change in how problems are defined and solutions evaluated, as summarized in Table 3. The old paradigm evaluated transport system performance based primarily on vehicle travel speeds using indicators such as roadway Level-Of-Service (LOS), traffic speeds and congestion delay. This approach is criticized for biasing planning in favor of automobile-oriented solutions (Roth 2009). The new paradigm evaluates performance based on overall accessibility and considers other planning objectives, impacts, and modes.

Table 3 Transport Planning Paradigms (ADB 2009; EA 2021; Litman 2013a)

	Old Paradigm	New Paradigm
Definition of Transportation	Mobility: movement of people and goods, particularly automobile travel.	Accessibility: people's ability to research desired services and activities.
Planning goals	Maximize motor vehicle travel speed and affordability.	Improve overall accessibility and transport system efficiency.
Modes considered	Automobile, truck and transit.	Multiple modes and transport services.
Performance indicators	Vehicle travel speeds, roadway Level-of-Service, and cost per person-mile.	Quality of transport options. Proximity of destinations. Per capita transport costs.
Favored transport improvements	Roadway and parking facility expansions. Vehicle improvements.	Multimodal improvements. Transportation demand management. Smart Growth policies.

This table compares the old and new transport planning paradigm.

Table 4 illustrates the scope of modes and impacts considered in planning. Conventional planning evaluates transport system performance based primarily on congestion intensity using indicators such as roadway Level-Of-Service (LOS) and the Travel Time Index (TTI). These only measure motor vehicle delay; they indicate nothing about other modes (rail transit and non-motorized modes) or other impacts (passenger comfort, parking costs, safety and security, etc.).

Table 4 **Conventional Analysis Scope**

		More Modes →		
		Automobile	Public Transit	Walking/Biking
← More Impacts	Travel speed	Auto Delay	Bus delay	
	Consumer costs and affordability	Fuel costs	Bus fuel costs	
	Travel convenience and comfort			
	Parking convenience and costs			
	Safety and security			
	Mobility for non-drivers			
	Pollution emissions			
	Public fitness and health			

Conventional transport evaluation considers a limited set of modes and impacts; automobile delay receives the most consideration (darkest blue cell), increased fuel costs and bus delay receive less consideration (lighter blue cells), and other impacts often receive little consideration. The new planning paradigm requires more comprehensive and multimodal analysis.

Recent research improves our understanding of factors that affect overall accessibility and the trade-offs between them. For example, Ewing and Cervero (2010) found that a 10% increase in roadway connectivity reduces average travel distances by 1.2%. Levine, et al. (2012) found that urban density affects motorists' access to destinations far more than traffic speeds. Owen, Murphy and Levinson (2018) found that accounting for peak-period traffic speeds (therefore congestion delays) and travel distances, denser cities such as Los Angeles, San Francisco, New York have better job access than more sprawled cities such as Dallas, Houston and Atlanta. Kuzmyak (2012) found that residents of more compact and multimodal neighborhoods experience less congestion delay than residents in automobile-dependent areas. Ng and Small (2012) find that slower urban roadways often carry more capacity than higher speed urban highways. Mondschein and Taylor (2017) found that "congestion-adapted" places tend to have fewer car trips but more total trips and better access overall (they can reach more services and activities). Ewing, et al. (2017) found that more compact development reduces but concentrates vehicle travel, which roughly cancel each other out, so by itself, increasing density typically has neutral impacts on congestion costs.

A Transportation for America report, *The Congestion Con: How More Lanes and More Money Equals More Traffic* (TfA 2020) analyzed how roadway expansions affected per capita congestion delay in the 100 largest urbanized areas in the U.S. between 1993 and 2017. During that period governments spent more than \$500 billion on highway projects but congestion grew by 144%, far more than population, and the regions that expanded their roads most had little success reducing congestion. The authors concluded that this resulted from generated traffic which filled the added capacity. Engineers Australia's *Urban Transport Systems* discussion paper concluded, "There is little evidence to demonstrate increasing road capacity to reduce traffic congestion improves economic performance in cities."

When evaluated based on *mobility*, transportation performance is measured based on travel speed, but when evaluated based on *accessibility*, transportation system performance is measured based on the total time and money costs required to research desired destinations. This recognizes the accessibility benefits of more connected transportation networks and more compact development, which reduce travel distances.

In a detailed study of travel activity in Halifax, Canada, Millward and Spinney (2011) found that the total amount of time people spend travelling declines with compact development, despite the fact that urban residents rely more on slower modes (walking, bicycling and public transit), and their automobile travel is slower. This indicates that travel distance, and therefore the dispersion of destinations, is more important than speed in determining total travel time costs. They found that total average time spent travelling increased from 92 daily minutes for urban residents, 94 daily minutes for suburban residents, 107 daily minutes for closer exurban and 104 for the most distant exurban residents. Mean one-way commute durations increased from 12.7 minutes in the inner city, 15.7 minutes in suburbs, 18.1 minutes for closer exurbs and 21.9 minutes for the most distant areas. Urbanites spend more time walking, bicycling, and using transit, and a smaller proportion of travel time in cars: inner-city respondents average only 56 minutes per day in a car (45 as driver, 11 as passenger), whereas suburbanites average 72 minutes, and exurban residents average 85 to 91 minutes. Average daily time devoted to active travel (walking and bicycling) declined from 27.8 in urban areas, 16.5 in suburbs, 13.7 for closer exurbs and 13.2 in outer exurbs.

New tools can help apply accessibility-based evaluation (Levinson and King 2020). For example, multimodal LOS can be used to rate walking, cycling and public transit service quality (Dowling, et al, 2008). Accessibility indicators and models measure the time and money required to reach destinations, such as the number of jobs or retail services available within a given travel time by various modes. These models take into account travel speeds, network connectivity and the distribution of destinations, as illustrated in Figure 3. Such analyses can be disaggregated to indicate accessibility for specific groups or trips, such as children's ability to walk and bicycle to school, low-income non-drivers' access to healthcare services and grocery stores, or the number of service jobs within reasonable travel time of adolescents' homes.

Figure 3 Multimodal Access Mapping (Slavin, Rabinowicz and Flammia 2013)

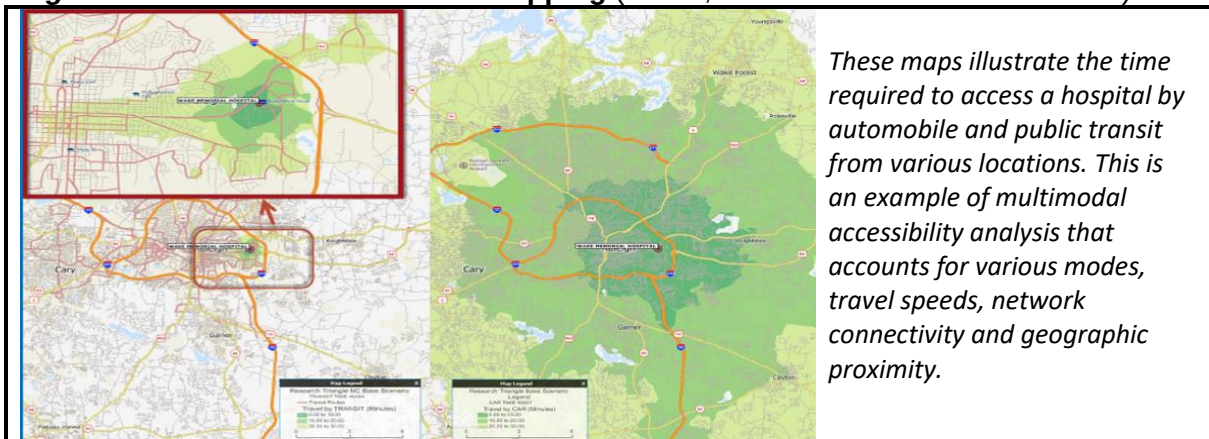


Table 5 summarizes various accessibility factors and compares their current evaluation practices with what is required for comprehensive and multimodal planning. For example, comprehensive evaluation recognizes that improving walking and cycling conditions, public transit comfort, roadway connectivity, development density and mix, and mobility substitutes such as telecommunications can all increase accessibility. Traffic-oriented indicators such as roadway LOS and the Travel Time Index ignore these factors.

Table 5 **Consideration of Accessibility Factors In Transport Planning**

Factor	Consideration in Conventional Evaluation	Required for Comprehensive Evaluation
Automobility – motor vehicle traffic speed, congestion delays, vehicle operating costs, crash rates per mile or kilometer	Usually considered using indicators such as roadway level-of-service, average traffic speeds and congestion costs and crash rates.	Impacts should be considered per capita (per capita vehicle costs and crash casualties) to take into account the amount that people travel.
Quality of other modes – convenience, comfort and safety of walking, cycling and transit	Considers public transit speed but not comfort. Non-motorized access is often ignored.	Multimodal performance indicators that account for convenience, comfort, safety, affordability and integration.
Transport network connectivity – density of connections between paths, roads and modes, and therefore the directness of travel	Traffic network models consider regional road and transit networks but often ignore local streets, non-motorized networks, and intermodal connections.	Fine-grained analysis of path and road network connectivity, and connections between modes, such as the ease of walking and biking to transit stations.
Land use accessibility – development density and mix, and therefore travel distances	Often ignored. Some integrated models consider some land use factors.	Fine-grained analysis of how land use factors affect accessibility by various modes.
Mobility substitutes –delivery services and telecommunications that reduce the need to travel	Only occasionally considered in conventional transport planning.	Consider these accessibility options in transport planning.

Conventional planning evaluates transport system performance based primarily on regional travel speed. Additional factors must be considered for comprehensive accessibility evaluation.

Accessibility-based planning expands the range of solutions that can be applied to congestion problems. The old, mobility-based paradigm assumed that traffic congestion reflected a shortage of roadway capacity and so considered roadway expansions the preferred solution. Accessibility-based planning compares roadway expansions with other possible solutions: shifts from peak to off-peak travel times, improving and encouraging space-efficient modes (walking, bicycling, ridesharing, public transit and telework), Smart Growth development policies that reduce the distances that people must drive, plus pricing reforms and new information technologies that support these shifts.

If there are enough vehicles travelling on a corridor to create congestion, there are enough travellers to justify frequent public transit and ridesharing services. Dedicated bus or HOV lanes are justified if, once completed those modes would carry more than 2,200 passengers on a grade-separated highway or 800 passengers on a surface street, since that is more travellers and therefore more time savings than a general traffic lane (Litman 2015a).

Quantifying and Monetizing Congestion Costs

Various methods are used to *quantify* (measure) and *monetize* (measure in monetary units) congestion costs. This section describes the methods recommended by experts (Grant-Muller and Laird 2007; OECD/ECMT 2007; TC 2006; Wallis and Lupton 2013).

Analysis Scope

Many factors may affect congestion, such as city size and density, changes in employment rates and business activity. For example, since both transit ridership and congestion intensity tend to increase with city size, density and employment rates, failing to account for these factors can lead to a false conclusion that increased transit ridership contributes to congestion.

Table 6 describes various congestion indicators. Some only measure vehicle traffic delay at a particular location, others are more comprehensive (they consider overall travel delay, speeds and distances) and multimodal (they consider delays to all travelers, not just motorists).

Table 6 Common Congestion Indicators (“Congestion Costs,” Litman 2009)

Indicator	Description	Comprehensive?	Multimodal
Roadway Level Of Service (LOS)	Congestion intensity at a particular location, rated from A (uncongested) to F (most congested).	No	No
Multimodal LOS	Congestion delays to various modes, rated from A to F.	No	Yes
Travel Time Index	The ratio of peak period to free-flow traffic speeds.	No	No
Average Traffic Speed	Average vehicle travel speeds at a particular location.	No	Yes if for all modes
Commute Duration	Average time per commute trip.	No	Yes if for all modes
Per Capita Travel Time	Total average time residents devote to travel.	Yes if for all modes	Yes if for all modes
Percent Travel Time In Congested Conditions	Portion of peak-period vehicle or person travel that occurs under congested conditions.	No	Yes if for all modes
Congestion Duration	Average duration of congested conditions.	No	No
Congested Lane Miles	Number of lane-miles congested during peak periods.	No	No
Annual Hours Of Delay	Hours of extra travel time due to congestion.	Yes if for all modes	Yes if for all modes
Annual Delay Per Capita	Hours of extra travel time divided by area population.	Yes if for all modes	Yes if for all modes
Excess Fuel Consumption	Total additional fuel consumption due to congestion.	No	No
Congestion Cost Per Capita	Hours of delay times monetized value of travel time, plus additional fuel costs, divided by area population.	Yes	Yes if for all modes
Planning Time Index	Earlier departure required during peak periods	No	No
Barrier Effect	Walking and cycling delay caused by wider roads	No	No

Congestion is evaluated using various indicators. Some are more comprehensive and multimodal than others.

Figure 4 illustrates examples of roadway *Levels of Service* (LOS), a widely-used indicator of congestion intensity which rates traffic conditions from A (freeflow) to F (highly congested).

Figure 4 Roadway Levels Of Service (HCM 2000, Ex. 21-3)

Multi-Lane Highway				Single-Lane Roadway			
Level of Service	Flow Conditions	Operating Speed (mph)	Technical Descriptions	Level of Service	Flow Conditions	Operating Speed (mph)	Technical Descriptions
A		60	Highest level of service. Traffic flows freely with little or no restrictions on maneuverability. No delays	A		55+	Highest quality of service. Free traffic flow with few restrictions on maneuverability or speed. No delays
B		60	Traffic flows freely, but drivers have slightly less freedom to maneuver. No delays	B		50	Stable traffic flow. Speed becoming slightly restricted. Low restriction on maneuverability. No delays
C		60	Density becomes noticeable with ability to maneuver limited by other vehicles. Minimal delays	C		45	Stable traffic flow, but less freedom to select speed, change lanes or pass. Minimal delays
D		57	Speed and ability to maneuver is severely restricted by increasing density of vehicles. Minimal delays	D		40	Traffic flow becoming unstable. Speeds subject to sudden change. Passing is difficult. Minimal delays
E		55	Unstable traffic flow. Speeds vary greatly and are unpredictable. Minimal delays	E		35	Unstable traffic flow. Speeds change quickly and maneuverability is low. Significant delays
F		<55	Traffic flow is unstable, with brief periods of movement followed by forced stops. Significant delays	F			Heavily congested traffic. Demand exceeds capacity and speeds vary greatly. Considerable delays

Two Way Stop Intersections				Intersections With Traffic Signals			
Level of Service	Flow Conditions	Delay per Vehicle (seconds)	Technical Descriptions	Level of Service	Delay per Vehicle (seconds)	Factors Affecting LOS of Signalized Intersections	
A		≤10	Very short delays	A		≤10	Traffic Signal Conditions: • Signal Coordination • Cycle Length • Protected left turn • Timing • Pre-timed or traffic activated signal • Etc. Geometric Conditions: • Left- and right-turn lanes • Number of lanes • Etc. Traffic Conditions: • Percent of truck traffic • Number of pedestrians • Etc.
B		11-15	Short delays	B		11-20	
C		16-25	Minimal delays	C		21-35	
D		26-35	Minimal delays	D		36-55	
E		36-50	Significant delays	E		56-80	
F		>50	Considerable delays	F		>80	

These images from the 2000 *Highway Capacity Manual* illustrate and describe various roadway levels of service. There are similar ratings for intersections (see www.dot.ca.gov/ser/forms.htm).

Baseline Speeds

A key congestion analysis factor is the *baseline* (also called *threshold*) speed. Comparing the baseline speed to the actual traffic speed determines the delay of a given area. For example, if the baseline speed is 60 miles per hour (mph), and actual traffic speeds are 50 mph, the delay is 10 mph. Baseline speeds can be defined in the following ways:

- *Free-flow speeds*: traffic speeds measured during uncongested conditions (LOS A).
- *Speed limits*: maximum legal speeds on a road (LOS A or B).
- *Capacity-maximizing speeds*: maximizes roadway vehicle traffic capacity (LOS C or D).
- *Efficiency-optimizing speeds*: reflects users' willingness-to-pay for faster travel (also called *consumer-surplus maximizing* or *deadweight loss minimizing*, usually LOS C or D).

As traffic speeds increase, so does the space required between vehicles (*shy distance*) for a given level of driver effort and safety. For example, a typical highway lane can efficiently carry more than 1,500 vehicles per hour at 45-54 mph, about twice the 700 vehicles that can operate comfortably at 60+ mph. Urban arterial capacity tends to peak at 35-45 mph. Maintaining freeflow speeds under urban-peak conditions is more costly than most motorists are willing to pay, and therefore economically inefficient. As a result, freeflow and speed limits are typically level-of-service (LOS) **A** or **B**, while capacity-maximizing and efficiency optimizing speeds are typically LOS **C** or **D** (Table 7).

Table 7 Typical Highway Level-of-Service (LOS) Ratings (TRB 2000)

LOS	Description	Speed (mph)	Flow (veh./hour/lane)	Density (veh./mile)
A	Traffic flows at or above posted speed limit. Motorists have complete mobility between lanes.	Over 60	Under 700	Under 12
B	Slightly congested, with some reduced maneuverability.	57-60	700-1,100	12-20
C	Ability to pass or change lanes constrained. Roads are close to capacity. Target LOS for most urban highways.	55-57	1,100-1,550	20-30
D	Speeds somewhat reduced, vehicle maneuverability limited. Typical urban peak-period highway conditions.	45-54	1,550-1,850	30-42
E	Irregular flow, speeds vary and rarely reach the posted limit. Considered a system failure.	30-45	1,850-2,200	42-67
F	Flow is forced, with frequent drops in speed to nearly zero mph. Travel time is unpredictable.	Under 30	Unstable	67-Maximum

This table summarizes roadway Level of Service (LOS) ratings, an indicator of congestion intensity.

Capacity-maximizing or efficiency-optimizing baseline speeds are considered an *economic* approach that maximizes efficiency and consumer benefits (Wallis and Lupton 2013). Most recent congestion cost studies use capacity-maximizing or economically efficient baseline speeds. For example, the Australian Bureau of Transport and Regional Economics recommends calculating congestion costs based on motorists willingness to pay for faster travel, described as, "the increase in net social benefit if appropriate traffic management or pricing schemes were introduced and optimal traffic levels were obtained" (BTRE 2007, p. 10). Using this method, they

estimate that Australian congestion costs totaled \$5.6 billion in 2005, half the \$11 billion calculated using freeflow speeds. Similarly, Wallis and Lupton (2013) estimate that the use of capacity optimizing speeds for Auckland reduced New Zealand congestion costs to \$250 million, a third of the \$1,250 million estimate based on freeflow speeds. Transport Canada calculates congestion costs use 50%, 60% and 70% of free-flow speeds, values that they consider a reasonable range of optimal urban-peak traffic speeds.

For these reasons, most transport economists recommend capacity-maximizing or economic efficiency-optimizing traffic speeds rather than freeflow baseline speeds (TC 2006; Wallis and Lupton 2013). One leading economist explains,

“The most widely quoted [congestion cost] studies may not be very useful for practical purposes, since they rely, essentially, on comparing the existing traffic conditions against a notional ‘base’ in which the traffic volumes are at the same high levels, but all vehicles are deemed to travel at completely congestion-free speeds. This situation could never exist in reality, nor (in my view) is it reasonable to encourage public opinion to imagine that this is an achievable aim of transport policy.” (Goodwin 2003)

Newer studies use baselines based on actual measured freeflow traffic speeds, which often exceed legal speed limits. For example, the *Urban Mobility Report* (TTI 2012) used a 64.6 mph freeflow baseline speed for Los Angeles freeways which have 55 mph speed limits, and a 64.0 mph baseline for Miami freeways that have 60 mph speed limits (Table 8). These values indicate that 55-60% of these areas estimated congestion “costs” consist of speed limit compliance. Uncongested traffic usually exceeds speed limits, which are typically set to reflect 85th percentile freeflow speeds. Assuming that Los Angeles and Miami represent the higher range, this indicates that between approximately a quarter and a half of the UMR’s estimated congestion costs consist of speed compliance.

Table 8 UMR Peak Versus Freeflow Speed Table (TTI 2012)

Urban Area	Freeway		Arterial Streets		Urban Area	Freeway		Arterial Streets	
	Peak Speed	Freeflow Speed	Peak Speed	Freeflow Speed		Peak Speed	Freeflow Speed	Peak Speed	Freeflow Speed
Very Large Areas					Large Areas				
Atlanta GA	56.5	64.7	36.3	44.1	Minneapolis-St. Paul MN	54.3	63.8	39.6	43.1
Boston MA-NH-RI	54.2	63.4	29.5	36.0	Nashville-Davidson TN	57.2	64.1	34.2	41.9
Chicago IL-IN	53.0	63.1	34.3	40.2	New Orleans LA	54.9	63.2	39.6	43.7
Dallas-Fort Worth-Arlington TX	54.0	64.1	33.1	39.1	Orlando FL	58.8	64.3	34.9	42.8
Detroit MI	57.0	64.3	33.4	38.7	Pittsburgh PA	55.2	62.6	33.3	40.1
Houston TX	54.2	63.6	33.9	40.2	Portland OR-WA	49.2	60.3	31.1	36.5
Los Angeles-Long Beach-Santa Ana CA	48.6	64.6	37.4	43.7	Providence RI-MA	56.1	61.9	30.9	35.0
Miami FL	56.7	64.0	31.7	39.2	Raleigh-Durham NC	61.3	64.1	39.1	45.4
New York-Newark NY-NJ-CT	52.0	62.2	31.9	40.5	Riverside-San Bernardino CA	54.4	64.7	37.5	43.1
Philadelphia PA-NJ-DE-MD	55.5	63.6	31.8	39.2	Sacramento CA	55.2	64.7	37.4	43.5
Phoenix AZ	57.4	64.2	34.7	40.1	San Antonio TX	57.2	62.9	35.0	39.4
San Diego CA	56.8	64.5	37.6	43.7	Salt Lake UT	60.3	64.4	33.6	39.2
San Francisco-Oakland CA	54.0	64.1	37.8	44.0	San Jose CA	57.1	64.0	34.6	40.4
Seattle WA	51.2	62.0	30.4	35.2	San Juan PR	54.5	64.7	39.5	46.1
Washington DC-VA-MD	49.4	62.0	32.9	40.1	St. Louis MO-IL	44.4	56.0	29.8	34.9
					Tampa-St. Petersburg FL	59.1	64.2	37.2	44.2
					Virginia Beach VA	56.1	62.9	35.1	41.5
Large Areas									
Austin TX	52.9	62.6	36.2	42.9					
Baltimore MD	53.3	62.7	31.8	38.6					
Buffalo NY	55.2	62.0	33.4	38.6					
Charlotte NC-SC	58.0	62.9	34.0	41.4					
Cincinnati OH-KY-IN	56.3	63.7	32.5	38.2					
Cleveland OH	56.8	62.8	29.6	34.6					
Columbus OH	57.6	64.1	31.1	37.3					
Denver-Aurora CO	50.9	62.3	32.1	38.0					
Indianapolis IN	55.4	63.0	34.6	40.1					
Jacksonville FL	58.9	63.4	37.4	43.3					
Kansas City MO-KS	57.6	62.7	33.9	37.5					
Las Vegas NV	57.4	64.6	33.7	39.8					
Louisville KY-IN	57.0	63.7	34.0	39.9					
Memphis TN-MS-AR	56.9	64.0	36.1	42.5					
Milwaukee WI	55.6	62.5	35.7	39.3					

The Urban Mobility Report’s freeflow traffic speeds often exceed legal speed limits. In many cases more than half of the estimated congestion “cost” consists simply of speed limit compliance.

Data Collection

The methods used to collect and filter traffic data can bias results. For example, Inrix and TomTom indices use traffic speed data collected from their subscribers. These subscribers typically drive more than average under congested conditions. As a result, they oversample congested roadways and exaggerate congestion costs for average motorists (Salmon 2012).

Travel Time Values

Another key congestion costing factor is the value assigned for travel time and delay. There is extensive literature on this subject ("Travel Time Costs," Litman 2009; Grant-Muller and Laird 2007; USDOT 2011). Most studies conclude that motorists are willing to pay, on average, 20-40% of their wage rate for travel time savings, but these values are highly variable. Congestion increases drivers' stress, and so tends to have high costs per hour. Some travelers, such as commercial vehicles and people with urgent errands, have high values of travel time (NCHRP 2006; Parsons Brinckerhoff 2013; USDOT 2011), but most motorists are price sensitive.

Many toll road projects fail to achieve their traffic and revenue projections because few travellers are willing to pay cost-recovery prices (Prozzi, et al. 2009; Williams-Derry 2012). Burris, et al. (2016) found that about 7% of Katy Freeway motorists were willing to pay the equivalent of about \$40 per hour of time saved, and on average motorists are only willing to pay \$1.96 to \$8.06 per hour, much lower than generally assumed in congestion costing studies.

This indicates that it would be economically inefficient to invest significant resources to expand roadways to accommodate the lower-value trips, but efficiency increases if high-value trips are able to outbid lower-value traffic for road space.

Some congestion reduction strategies shift travel from automobiles to slower modes, such as walking, bicycling or public transit. There are debates concerning how to value this additional travel time. Travellers will sometimes choose slower modes, such as walking, bicycling and public transit in order to save money, exercise, relax or work while travelling (Smith, Veryard and Kilvington 2009). If travellers voluntarily shift from a faster to a slower mode in response to positive incentives (the slower mode has become more convenient or comfortable to use, or they receive a financial reward), they must be directly better off overall (an increase in overall consumer welfare) or they would not shift (Standen 2018). Conversely, if travelers shift their mode in response to negative incentives (such as increased user charges), they are probably directly worse off, although their overall benefits can depend on indirect impacts.

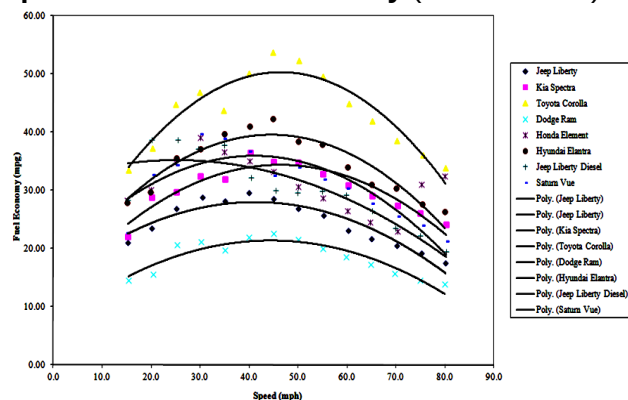
Perspective

Congestion evaluation can reflect various perspectives. Most congestion cost studies measure the costs that motorists *bear*, but for some applications, such as efficient road pricing or mode shift analysis it is important to calculate the marginal congestion costs a traveler *imposes* on others. These are generally higher than average values (Hau 1998). For example, when a road approaches its capacity, an additional vehicle may bear five minutes of delay but impose fifteen minutes of delay on other road users. Therefore, the additional vehicle's marginal congestion cost imposed is three times higher than the average congestion cost it bears. Similarly, if a three passenger car equivalent (PCEs) bus averages 30 passengers during peak periods, each passenger imposes one tenth the congestion cost of a car driver, and an additional passenger filling an otherwise unoccupied bus seat imposes virtually no marginal congestion cost.

Fuel Consumption and Emission Impacts

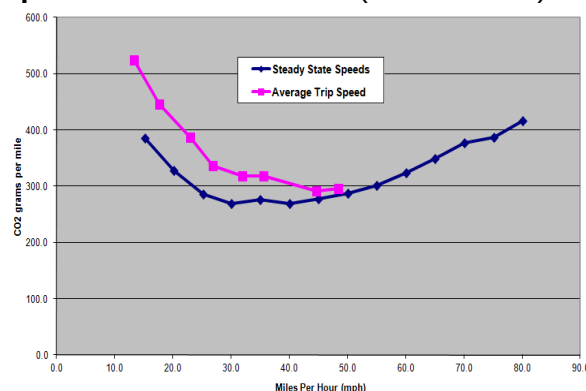
Other important factors are the formulas used to calculate how traffic speeds affect fuel consumption and pollution emissions. These are generally minimized at 40-50 miles per hour (mph), and increase above 55 mph (Barth and Boriboonsomin 2009; Bigazzi and Figliozzi 2012; ORNL 2012, Table 4.28), as illustrated in figures 5-A and 5-B.

Figure 5-A
Speed Versus Fuel Economy (USEPA Data)



USEPA data indicate that vehicle fuel economy tends to peak at 40-50 mph and declines above 55 mph.

Figure 5-B
Speed Versus Emissions (USEPA Data)



USEPA data indicate that average emission rates tends to increase above about 50 mph.

In addition, some congestion reduction strategies, such as roadway expansions, induce additional vehicle travel, which increases total fuel consumption and emissions, while others, such as improvements to resource-efficient modes, efficient transport pricing, and more accessible land use development, tend to reduce per capita vehicle travel and therefore total emissions, regardless of how they affect per-mile emission rates.

Safety Impacts

Although crash rates tend to increase with traffic density (vehicles per lane-mile), crash casualties (injuries and deaths) tend to decline if congestion significantly reduces traffic speeds (Kockelman 2011). Total crash rates tend to be lowest on moderately congested roads ($V/C=0.6$) and increase at lower and higher congestion levels (Marchesini and Weijermars 2010). Casualty rates (injuries and deaths) often increase when congestion is eliminated (Potts, et al. 2014; Zhou and Sisiopiku 1997). For example, using the TomTom Traffic Index (TomTom 2014), the five most congested U.S. cities (Los Angeles, San Francisco, Honolulu, Seattle and San Jose) average 5.6 traffic deaths per 100,000 residents, about half the 10.2 fatality rate of the ten least congested cities (Richmond, Birmingham, Cleveland, Indianapolis and Kansas City).

Per capita traffic deaths tend to increase with per capita vehicle travel, so roadway expansions that induce additional vehicle travel tend to increase traffic casualties (Luoma and Sivak 2012). One study estimated that the increased crash costs that result from reduced congestion offset 5-10% of congestion reduction benefits (Wallis and Lupton 2013).

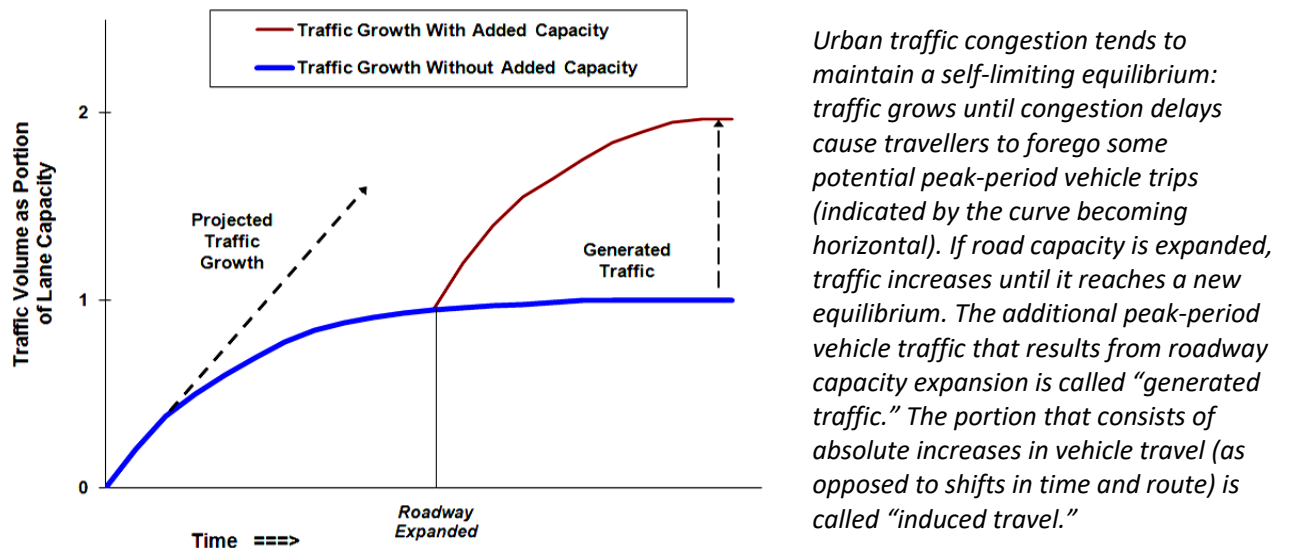
New Technologies

New telecommunications technologies and services (mobile phones, navigation devices, traffic information services, etc.) can help travellers avoid traffic congestion. By increasing total vehicle trips, ride hailing services are increasing urban traffic congestion (Schaller 2017). Some people claim that autonomous vehicles will reduce congestion by allowing vehicles to drive closer together in platoons, but this is only possible on grade-separated highways where such vehicles have dedicated lanes (Litman 2018).

Generated Traffic and Induced Travel

Traffic congestion tends to maintain equilibrium, it increases until delays cause some travelers to reduce their peak-period vehicle trips by shifting travel times, routes, modes and destinations (Arnott 2013; Hymel 2019; Jaffe 2014). If roads are expanded, traffic volumes will increase until congestion once again constrains peak-period trips, as illustrated in Figure 6. The additional peak-period vehicle travel on an expanded roadway is called *generated traffic*, and net increases in total vehicle travel are called *induced travel*.

Figure 6 How Road Capacity Expansion Generates Traffic (Litman 2001)



This has the following implications for congestion evaluation (Handy 2015; Hymel 2019; Kim 2022; Litman 2001; Næss, Nicolaisen and Strand 2012; Tennøy, Tønnesen and Gundersen 2019):

- Traffic congestion seldom becomes as severe as may be predicted by extrapolating past trends. As congestion increases, it discourages further peak-period trips.
- Roadway expansion provides less long-term congestion reduction benefits than predicted if generated traffic is ignored.
- Induced vehicle travel increases various external costs, including downstream congestion, parking costs, accident risk, and pollution emissions, reducing net benefits.
- Induced vehicle travel directly benefits the people who increase their vehicle travel. These benefits tend to be modest because the additional travel consists of marginal-value vehicle mileage that users are most willing to forego if their costs increase.

Congestion Cost Evaluation

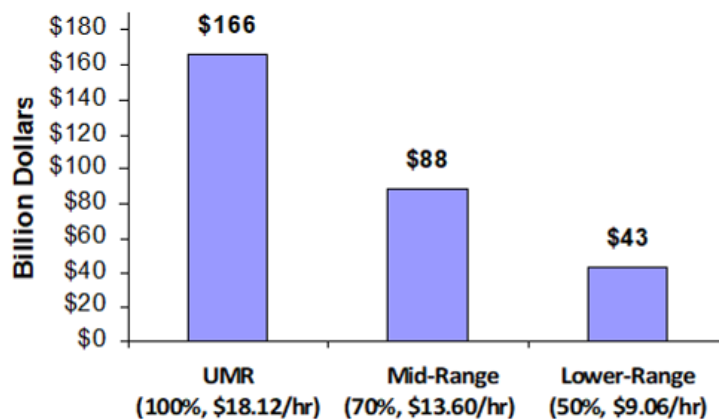
This section summarizes various monetized estimates of congestion costs, and compares congestion with other costs of transportation.

Congestion Cost Estimates

Various studies have monetized congestion costs for particular areas:

- INRIX (2019) claims that congestion costs “each American” 97 hours of additional travel time worth an estimated \$1,348 annually, but these figures really refer to approximately 15-25% of all Americans who are urban-peak auto commuters, and they use freeflow baseline speeds and value travel time at full wage rates.
- The Texas Transportation Institute’s *Urban Mobility Study* (the results of which are incorporated into various documents, such as the USDOT’s annual *Conditions & Performance* report) estimated that U.S. congestion cost totaled \$166 billion in 2017 (TTI 2019), and by extrapolating past trends it predicts that these costs will increase to \$199 billion in 2020. These values represent upper-bound estimates, since they are based on freeflow baseline speeds, higher than recommended travel time costs, optimistic fuel saving and emission reduction impacts, and no consideration of induced travel impacts (Cortright 2011; Litman 2019). More realistic assumptions result in lower estimates (Figure 7).

Figure 7 Congestion Cost Ranges (Litman 2019)



The Urban Mobility Report’s \$166 billion cost estimate is based on higher baseline speeds and travel time unit costs than most experts recommend. The lower-range estimate in this graph is based on 50% of baseline speed and the U.S. Department of Transportation’s lower travel time unit costs, reflecting reasonable lower-bound values.

- Dachis (2013) argues that conventional analysis underestimates total congestion costs by ignoring the negative effect it has on labor access. He concludes that including these impacts would increase monetized congestion costs by 25-85%.
- The American Association of Highway and Transportation Officials *Bottom Line* report (AASHTO 2014) estimates that if U.S. annual vehicle travel grows at 1.4% annually it must spend \$144 billion for roadway expansion, repair and maintenance, but if vehicle travel only grows 1.0% annually, required expenditures decline to \$120 billion. This suggests that 0.4% vehicle travel growth, about 12 billion VMT, causes \$24 billion in annual roadway costs.
- Kim (2019) estimated that US commuter congestion costs total about \$29 billion in 2010. A detailed study titled, “Does Roadwork Improve Road Speed?” (Kim 2022) concluded that the congestion-relieving effect of urban roadway expansions is temporary and generally insufficient to justify the large expenditures required.

- Transport Canada research calculated congestion costs using various roadway speed baselines (TC 2006), as summarized in Table 9.

Table 9 Total Costs of Congestion (TC 2006, Table 5)

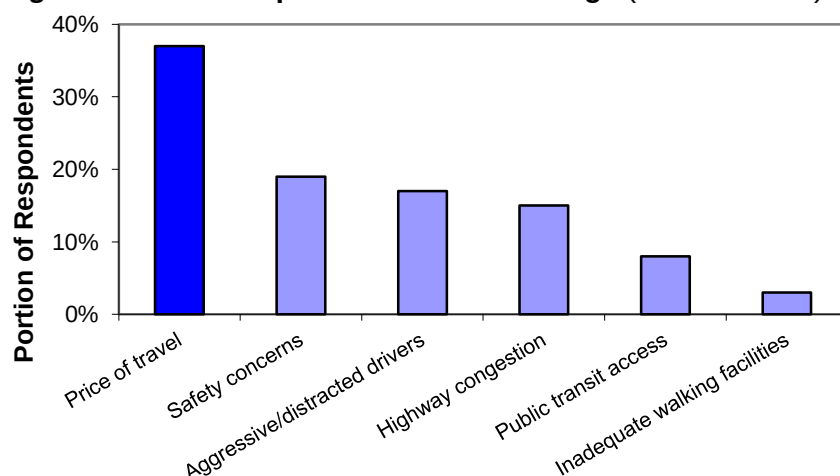
City	Relative To Freeflow Speeds		
	50%	60%	70%
Vancouver	\$403	\$517	\$629
Edmonton	\$49	\$62	\$74
Calgary	\$95	\$112	\$121
Winnipeg	\$48	\$77	\$104
Hamilton	\$6.6	\$11	\$17
Toronto	\$890	\$1,267	\$1,632
Ottawa-Gatineau	\$40	\$62	\$89
Montreal	\$702	\$854	\$987
Quebec City	\$38	\$52	\$68
Totals	\$2,270	\$3,015	\$3,721

Transport Canada calculates congestion costs based on 50%, 60% and 70% of freeflow speeds, which they consider the economically optimal range of urban-peak traffic speeds.

Congestion Compared with Other Costs

It is helpful to compare congestion with other urban transportation costs. Several studies have monetized various transport costs (CE, INFRAS, ISI 2011; Litman 2009; TC 2005-08). For example, the 2009 U.S. National Household Travel Survey asked respondents to rank various transport problems; the results indicate that transport system users consider congestion a moderate-priority problem, less important than financial costs or traffic safety (Figure 8).

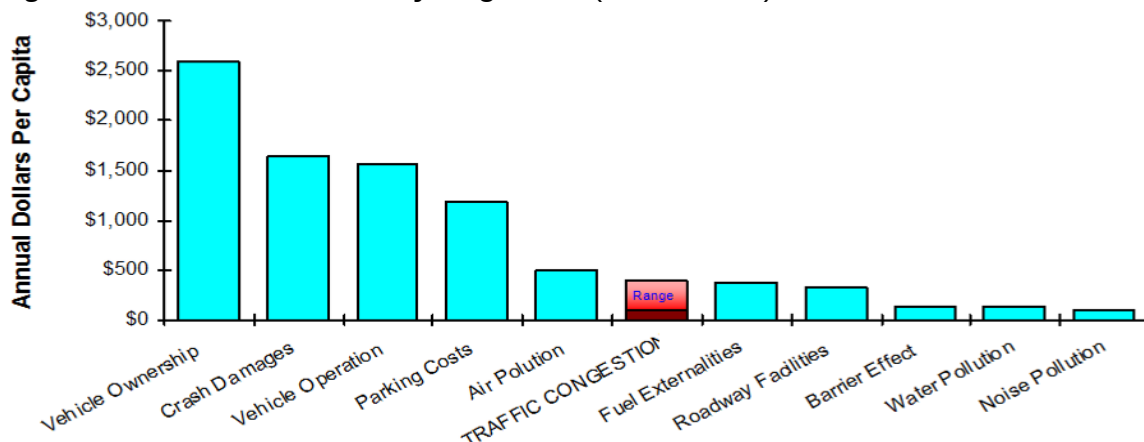
Figure 8 Transportation Issues Ratings (Mattson 2012)



National Household Travel Survey respondents ranked traffic congestion a moderate problem, far less important than financial costs or traffic safety.

Compared with other transportation costs, congestion costs are moderate, larger than some but smaller than others. For example, U.S. congestion costs are estimated to range between \$110 and \$390 annual per capita (Litman 2019; TTI 2019). This value can be compared with about \$4,000 in vehicle costs, \$1,500 in crash damages, \$1,000 in parking costs, \$500 in air and noise pollution costs and \$325 in roadway costs, as illustrated in Figure 9.

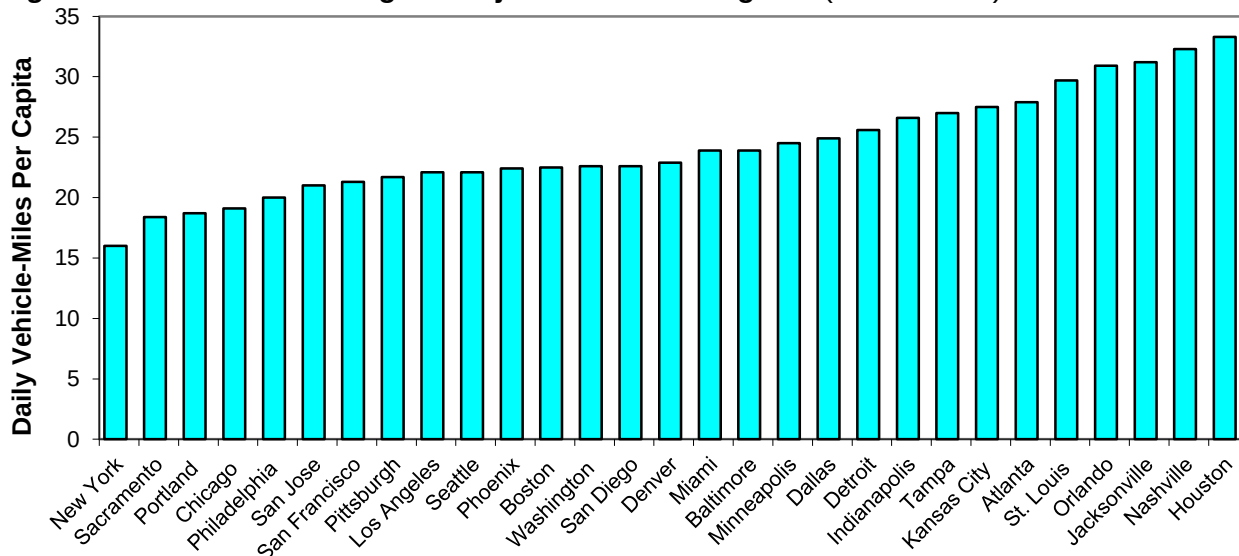
Figure 9 Costs Ranked by Magnitude (Litman 2009)



Congestion cost estimates range between \$110 and \$390 annually per capita, depending on assumptions. Even the highest estimate is moderate compared with other transport costs.

It is also useful to compare congestion with other factors that affect travel time and money costs. For example, the *Urban Mobility Report* indicates that in large cities, congestion costs auto commuters an additional 38 hours and 19 gallons of fuel annually. Sprawled, automobile-dependent development also increases travel time and fuel costs (Cortright 2010; Ewing and Hamidi 2014). For example, residents of sprawled communities such as Jacksonville, Nashville and Houston drive almost twice the daily miles as residents of more compact, multimodal regions such as New York, Sacramento and Portland (Figure 10). This additional vehicle travel requires about 104 additional hours and 183 additional gallons of fuel annually per resident (assuming 35 miles per hour and 20 miles per gallon averages). This suggests that sprawl imposes more than three times as much incremental transportation costs as congestion.

Figure 10 Vehicle Mileage in Major U.S. Urban Regions (FHWA 2008)



Per capita vehicle mileage varies significantly between U.S. urban regions.

Cortright (2010b) found that residents in more compact cities tend to spend less time in peak hour traffic due to shorter trips. In the best performing cities, those with the shortest peak hour travel distances, such as Chicago, Portland and Sacramento, the typical traveler spends 40 fewer hours per year in peak hour travel than the average American. In contrast, in the most sprawling metropolitan areas, such as Nashville, Indianapolis and Raleigh, the average resident spends as much as 240 hours per year in peak period travel because travel distances are so much greater. Cortright (2017) found that residents of urban regions with higher average traffic speeds but longer average travel distances are less satisfied with their transportation systems than residents of more compact communities with slower speeds. This suggests that congestion intensity is less important than total delay hours or total time spent driving.

Automobile-dependent communities also require more *chauffeuring* (also called *escort trips*), which refers to special vehicle trips to transport a passenger (Litman 2015b). Drivers' chauffeuring burden can be estimated by multiplying the ratio of non-drivers to drivers, times non-drivers' trip generation rates, times the portion of these trips that require chauffeuring, times their average trip duration, times two (for empty backhauls). A typical driver in an automobile-oriented community spends an additional 44 hours and 67 gallons of fuel chauffeuring non-drivers in their household. The results suggest that in automobile-dependent communities chauffeuring time and money costs are generally greater than congestion costs.

A congestion reduction strategy that increases other transport costs provides less total benefits, while a strategy that reduces other costs provides more total benefits, than indicated by analysis that only considers congestion impacts. For example, if a roadway expansion reduces congestion by 20%, but induces additional vehicle travel that increases parking, accident and pollution costs, the congestion reduction benefits are offset by other cost increases. However, if a public transit improvement or pricing reform reduces congestion by 10% and other costs by 5% each, total benefits are far larger, as illustrated in Table 10.

Table 10 Cost Analysis Example (APC = Annual Per Capita)

Cost Category	Current	Roadway Expansion		Improve Alt. Modes	
	APC Dollars	APC Dollars	Change	APC Dollars	Change
Congestion costs (mid-value)	\$250	\$200	-20%	\$225	-10%
Vehicle costs	\$4,000	\$4,200	+5%	\$3,800	-5%
Crash damages	\$1,500	\$1,575	+5%	\$1,425	-5%
Parking costs	\$1,000	\$1,050	+5%	\$950	-5%
Air and noise pollution costs	\$500	\$525	+5%	\$475	-5%
Roadway facility costs	\$325	\$341	+5%	\$309	-5%
Totals	\$7,575	\$7,891	+4.2%	\$7,184	-5.2%

In this example, per capita transport costs currently total \$7,575. A roadway expansion that reduces congestion 20% but increases other costs 5% increases total costs 4.2% to \$7,891. Alternative mode improvements that reduce congestion 10% and other costs 5% reduces total costs 5.2% to \$7,184.

Other potential congestion reduction strategies involve similar tradeoffs. For example, one concept is to divide urban highway lanes in two; this would allow accommodation of more motorcycles and half-width commuter vehicles (Figure 11). This strategy could reduce congestion but probably increase vehicle ownership (most users would need to acquire skinny vehicles in addition to general-purpose automobiles), residential parking and accident costs.

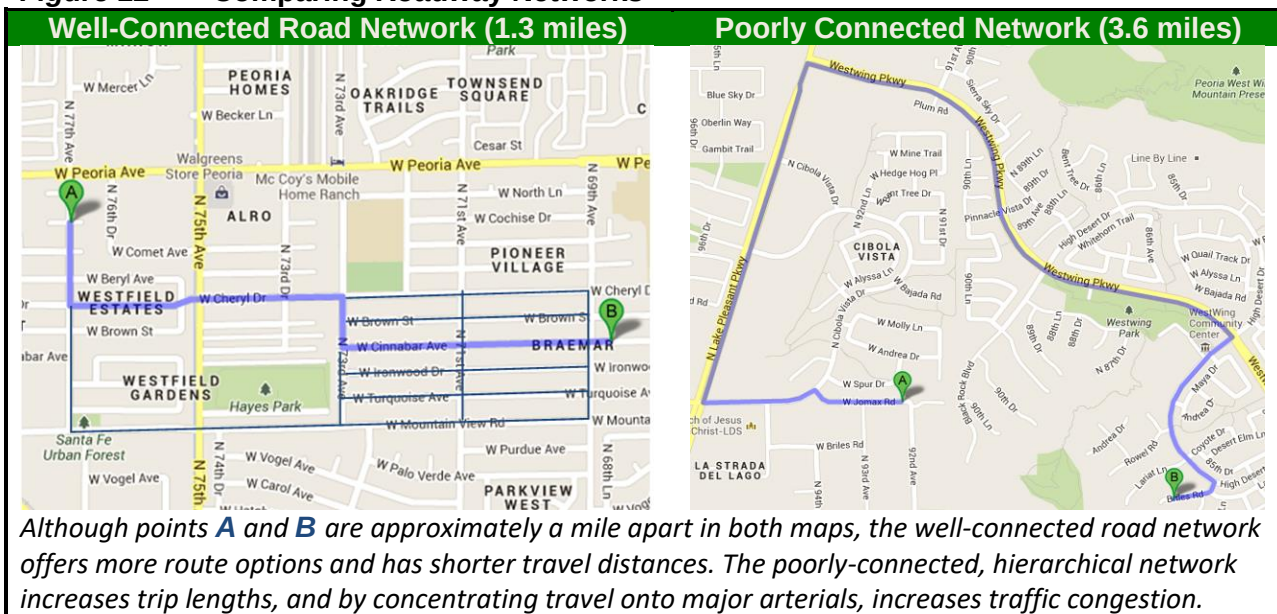
Figure 11 Half-Width Commuter Vehicles (www.commutercars.com)



Motorcycles and half-width commuter vehicles are sometimes proposed as a congestion reduction strategy. Under optimal conditions they can double the maximum number of vehicles per highway lane, but they are usually owned in addition to a general purpose vehicle, and so tend to increase vehicle ownership, residential parking and accident costs.

Another strategy involves the use of a hierarchical road network with fewer roadway intersections and more one-way arterials. This planning concept may increase traffic speeds but reduce roadway connectivity and concentrate more traffic on major roadways, which reduces accessibility and increases travel distances (Figure 12). Wider roads, hierarchical roadway networks and sprawled development patterns may appear attractive if evaluated using conventional congestion indicators since they increase traffic speeds. However, they may not be justified if evaluated using more comprehensive and multimodal performance indicators which consider impacts on all modes and all accessibility factors. Comprehensive and multimodal performance indicators account for indirect costs such as those associated with increased vehicle travel and automobile dependency.

Figure 12 Comparing Roadway Networks



Guidelines for Comprehensive and Multimodal Congestion Evaluation

This section describes factors that should be considered in comprehensive and multimodal congestion evaluation. For more discussion see Grant, et al (2011) and OECD/ECMT (2007).

Accessibility Analysis

Comprehensive and multimodal evaluation considers various accessibility factors, and therefore, trade-offs between them. Accessibility factors include:

- *Automobility* – motor vehicle traffic speed, congestion delays, affordability, and crash rates per mile or kilometer.
- *Quality of other modes* – speed, convenience, comfort, safety and affordability of walking, cycling, public transport and other modes.
- *Transport network connectivity* – density of connections between paths, roads and modes, and therefore the directness of travel between destinations.
- *Land use accessibility* – development density and mix, and therefore, travel distances.
- *Mobility substitutes* – telecommunications and delivery services that substitute for mobility.

Table 11 illustrates how various congestion reduction strategies affect accessibility factors. For example, roadway expansions tend to reduce walking and cycling access, directly by creating barriers to their movement, and indirectly by dispersing development. Dispersed development patterns increase trip distances beyond convenient walking distances. Conversely, improving space-efficient modes, increasing connectivity and utilizing more compact development tend to increase accessibility in ways that do not increase mobility. These improvements are not necessarily recognized by indicators such as average traffic speed or roadway level-of-service.

Table 11 Congestion Reduction Impacts on Accessibility Factors

Accessibility Factors	Roadway Expansion	Improve Alt. Modes	Efficient Pricing	Smart Growth	TDM Programs
Automobile access	+	+/-	+/-	+/-	+/-
Walking & cycling access	-	+	+	+	+
Public transport	+ (bus)	+	+	+	+
Network connectivity	-	+		+	+/-
Land use accessibility	-	+	+	+	+
Mobility substitutes					+

(+ improves that access factor; - degrades that access factor) Roadway expansions increase automobile access but by degrading walking conditions, encouraging more dispersed development and reducing other forms of access. Improving space-efficient modes, pricing reforms and Smart Growth policies may reduce automobile access but improve access in other ways.

Comprehensive Impact Analysis

Comprehensive evaluation considers all significant *impacts* (benefits and costs), and *planning objectives* (specific things a community wants to achieve). Such analysis can be *qualitative* (described), *quantitative* (measured), or *monetized* (valued in monetary units) (DfT 2006; Litman 2003; NZTA 2010). Table 12 illustrates a qualitative analysis of how five congestion reduction strategies affect ten planning objectives. Of course, actual impacts will vary depending on various factors, so this analysis should be adjusted to reflect specific conditions.

Table 12 Qualitative Evaluation of Potential Congestion Reduction Strategies

Planning Objectives	Roadway Expansion	Improve Alt. Modes	Efficient Pricing	Smart Growth
Congestion reduction	Large short-term but declines	Small short-term but increases	Potentially large	Reduces traffic speeds but improves access options and reduces travel distances
Roadway cost savings	Increases roadway costs	Usually reduces total roadway costs	Usually reduces total roadway costs	Usually reduces total roadway costs
Parking savings	Increases costs	Reduces parking costs	Reduces costs	Reduces parking demand but may increase facility costs
Consumer savings and affordability	Mixed	Can provide large savings	Increases driving costs but provides other savings	Tends to reduce per capita transport expenditures
Improved non-driver access	Degrades walking conditions	Usually large benefits	Generally improves non-drivers' access	Large benefits
Improved traffic safety	Reduced crash rates offset by higher speeds and more vehicle travel	Usually increases safety	Usually increases safety	Usually increases safety
Reduced pollution	Reduced emission rates offset by more vehicle travel	Tends to reduce emissions	Tends to reduce emissions	Reduces emissions but may increase exposure to local pollutants
Energy conservation	Reduced fuel consumption rates but increased vehicle travel	Generally reduces per capita energy consumption	Generally reduces per capita energy consumption	Generally reduces per capita energy consumption
Efficient land use	Often causes sprawl	Supports more compact development	Supports more compact development	Supports more compact development
Improved fitness and health	Tends to reduce active transport	Usually increases active transport	Usually increases active transport	Usually increases active transport

Roadway expansion helps reduce congestion but tends to contradict other objectives. Other types of congestion reduction strategies tend to achieve more objectives.

Table 13 illustrates a quantitative evaluation of potential congestion reduction strategies' impacts. These strategies are rated from 3 (most positive) to -3 (most negative). These objectives can be weighted. For example, improved safety can be given twice the weight as energy savings, or vice versa (Litman 2003).

Table 13 Quantitative Evaluation of Potential Congestion Reduction Strategies

Planning Objectives	Roadway Expansion	Improve Alt. Modes	Efficient Pricing	Smart Growth
Congestion reduction	3	2	3	0
Roadway cost savings	-3	2	3	3
Parking savings	-2	2	3	3
Consumer savings and affordability	1	1	0	0
Improved access for non-drivers	-2	3	3	3
Improved traffic safety	-2	2	3	3
Reduced pollution	-2	2	3	3
Energy conservation	-2	2	3	3
Efficient land use	-3	2	3	3
Improved fitness and health	-3	3	3	3
Totals	-15	21	27	24

This quantitative analysis rates each strategy's impacts on ten planning objectives from 3 to -3.

Many of these impacts can be *monetized* (Litman 2009; TC 2005-08). Table 14 illustrates an example of a monetized evaluation of congestion reductions on an urban roadway with one million annual peak-period vehicle-miles. Both roadway expansion and transport demand management (TDM) strategies (a combination of improving space-efficient modes, efficient pricing, Smart Growth policies, and targeted programs) are assumed to reduce congestion 33%. Contrarily, roadway expansions would increase affected vehicle travel 10%, while the TDM strategies would reduce vehicle travel 10%. Both strategies provide \$45,000 annual congestion cost savings, but the roadway expansion benefits are largely offset by the additional costs of the induced travel. TDM strategies provide additional benefits (reduced road and parking costs, crashes, barrier effects, pollution and petroleum externalities, plus consumer savings from improved transport options) which approximately double the congestion reduction benefits.

Table 14 Monetized Evaluation of Potential Congestion Reduction Strategies

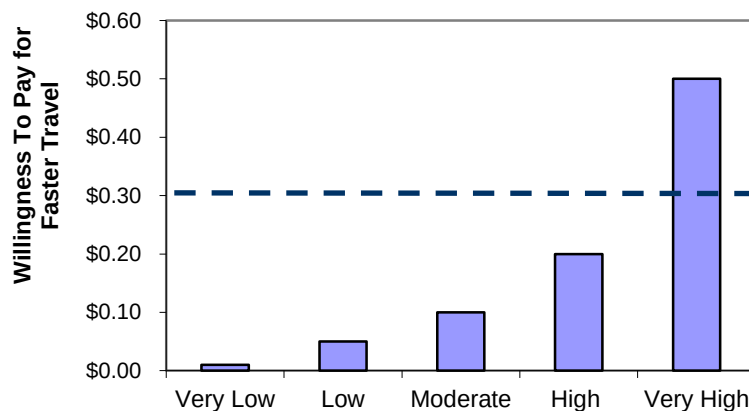
Costs	Costs per Veh.-Mile	Roadway Expansion	Transport Demand Management
Vehicle Travel Change		+10%	-10%
Congestion Costs		-33%	-33%
Congestion	\$0.15	-\$45,000	-\$45,000
Roadway operations	\$0.04	\$4,000	-\$4,000
Parking subsidies	\$0.10	\$10,000	-\$10,000
Vehicle operation	\$0.15	\$15,000	-\$15,000
Crash damages	\$0.10	\$10,000	-\$10,000
Barrier effect (pedestrian/cycling delays)	\$0.03	\$3,000	\$3,000
Air and noise pollution	\$0.05	\$5,000	-\$5,000
Petroleum externalities	\$0.02	\$2,000	-\$2,000
Totals		\$4,000	-\$88,000

Monetized analysis uses estimates of costs and benefits to calculate the value of a policy or project.

Economic Efficiency and Consumer Surplus

Economic efficiency recognizes the diversity of value provided by travel activity: some vehicles (urgent errands, freight trucks and buses carrying numerous passengers) tend to have high travel time values, while others have low value. Users would shift modes, routes or destinations if their costs increased or alternatives improved modestly. Figure 13 illustrates this concept, showing motorists' demand curve for faster travel. If expanding roadways to reduce congestion delays costs 30¢ per peak-period vehicle-mile, it is economically efficient to offer this option to the motorists whose willingness to pay exceeds this amount. Serving this demand reflects consumer sovereignty (the principle that consumer preferences should ultimately determine which goods and services are produced) and increases consumer-surplus (net user benefits). Contrarily, it is economically inefficient to spend that amount to increase the travel speeds of motorists with lower willingness-to-pay. Expanding roads to accommodate lower-value peak-period vehicle travel means that society is spending two dollars to provide a benefit that consumers only value at one dollar, and the expansion is particularly harmful if the added capacity induces additional vehicle travel which increases external costs.

Figure 13 **Faster Traffic Demand Curve**



On a typical road, users' willingness-to-pay for faster travel varies from very low to very high. If expanding urban roadways cost 30¢ per peak-period vehicle-mile, economic efficiency increases if motorists willing to pay this amount can purchase faster travel, but it would be economically inefficient to spend this amount to increase the travel speeds of motorists with lower willingness-to-pay.

This has various implications for congestion evaluation:

- There are large potential benefits of favoring higher-value travel. A roadway becomes more efficient (it provides more value per lane or vehicle-mile) if regulations, pricing or incentives allow higher value vehicles to avoid congestion.
- A significant portion of motor vehicle travel may have negative net value: its marginal user benefits are less than their total marginal costs. These marginal costs include external costs such as roadway costs, parking facility costs and accident and pollution damages. It is economically inefficient to expand roads to accommodate such travel.
- Improving transport options (alternatives to driving) that serve latent demand also reflects consumer sovereignty, and increases consumer-surplus. For example, walking, cycling and transit improvements that increase the use of those modes provide direct user benefits in addition to any indirect benefits from reduced automobile travel.
- Improving traveler convenience and comfort can reduce travel time unit costs (dollars per hour). These reductions are of equivalent value to those acquired from increasing travel speed.

Experience indicates that a 1¢ per vehicle-mile road toll typically reduces congested vehicle travel by about 1%, with larger reductions on urban highways that have good travel alternatives such as high quality public transport (PSRC 2008; Spears, Boarnet and Handy 2010). This reduction reflects the value that motorists place on their vehicle travel. For example, about 20% of peak-period motorists value their trips at less than 20¢ per vehicle-mile, and 30% value it less than 30¢; if charged those amounts, they would prefer to shift time, mode or destinations. Urban-peak travel has external costs (crashes, pollution, parking subsidies, barrier effect, petroleum externalities, etc.) of 20-30¢ per mile, and expanding urban roadways typically costs \$0.50 to \$1.50 per additional urban-peak trip accommodated (Litman 2009; TC 2005-08). As a result, a significant portion of urban-peak vehicle travel is probably worth less than its total cost. Much of the additional vehicle travel associated with the expansion of urban roadways is probably worth less than its total marginal costs (roadway expansion and other external costs).

The potential benefits of policies that favor higher-value vehicle travel are likely substantial. For example, freight, commercial and public transit vehicles often have values of time an order of magnitude higher than that of an average automobile, so giving them priority in traffic can provide large efficiency gains; their benefits more than offset losses to lower-value travelers.

Conventional congestion analysis generally ignores these issues. It seldom quantifies the economic efficiency gains of favoring higher value travel, the consumer surplus gains of serving latent demand, or the economic inefficiencies that result if roadway expansions induce additional vehicle travel that has marginal benefits worth less than marginal external costs. Comprehensive evaluation considers these factors.

Social Equity

Equity refers to the distribution of costs and benefits and whether they are considered fair and appropriate (Litman 2022). There are three general categories of equity to consider:

1. *Horizontal equity* (also called *fairness*) is concerned with whether similar people and groups are treated similarly. It suggests that people with comparable incomes and needs should receive similar shares of public resources and bear similar costs. It implies that users should “get what they pay for and pay for what they get” unless subsidies are specifically justified.
2. *Vertical equity with regard to income* considers the allocation of impacts between different income classes, assuming that policies should favor lower income people. Policies that provide a proportionally greater benefit to lower-income groups are called *progressive*, while those that make lower-income people relatively worse off are called *regressive*.
3. *Vertical equity with regard to mobility needs* considers whether a transport system provides adequate service to people with mobility impairments and other special needs. This type of equity justifies universal design (facilities designed to accommodate all users, including people with impairments) and policies that provide basic mobility to disadvantaged people (such as bus services), even if this requires subsidies.

Conventional congestion evaluation tends to consider a limited set of equity issues, such as whether congestion reduction funds are fairly allocated among different jurisdictions and whether decongestion pricing is regressive. These equity impacts tend to be overlooked:

- The unfairness of large public expenditures on highway expansions that benefit peak-period motorists. This justifies toll funding of highway expansions, and more investments in non-auto modes, so non-drivers receive their share of public investments.
- The inequity of higher-occupant vehicle (bus and carpool) passengers being delayed by traffic congestion caused by lower-occupant vehicle passengers who require 10 to 100 times more road space. This justifies bus and HOV lanes.
- The inequity of reduced pedestrian and cycling safety and accessibility caused by wider roads, increased traffic speeds, reduced roadway connectivity and sprawled development. This indicates that there is an equity justification for favoring narrower roads, lower traffic speeds, and other pedestrian and cycling improvements.
- The regressivity of congestion reduction strategies that favor automobile travel over more affordable modes (walking, cycling and public transport) and therefore forces lower-income households to own more vehicles than they can afford.
- The harm that automobile-dependency and sprawl have on physically, economically and socially disadvantaged people.

Comprehensive equity analysis tends to support congestion reduction strategies that improve affordable modes (walking, cycling and public transport), decongestion pricing (higher tolls under congested conditions), TDM incentives that reward travellers who use space-efficient modes, and Smart Growth policies that create more multimodal communities where it is easier to get around without driving (Manville 2017; Shaheen, Stocker and Meza 2020).

Comprehensive Evaluation Summary

Table 15 summarizes the major factors that should be considered in comprehensive and multimodal congestion evaluation framework.

Table 15 Comprehensive Congestion Evaluation Framework

	Accessibility Factors	Impacts	Economic Efficiency	Social Equity
Major factors to consider in comprehensive evaluation	• Automobile accessibility • Accessibility by other modes • Roadway connectivity • Geographic proximity (land use density and mix)	• Traffic congestion • Road and parking costs • Accidents • Consumer costs • Mobility for non-drivers • Energy consumption • Pollution emission • Efficient land use • Public fitness	• Efficiency gains from favoring higher value trips • Possibility that induced travel has negative net benefit • Consumer surplus gains from new modes and services	• Fairness of benefit and costs allocation. • Impacts on physically, economically and socially disadvantaged people. • Unfairness of policies that favor automobile travel over other modes
Conventional congestion evaluation	Primarily considers automobile access; other accessibility factors are often overlooked.	Focuses on travel speed and vehicle operating costs. Other impacts are often overlooked or undervalued.	Generally ignores economic efficiency factors.	Generally considers a limited set of equity impacts.
Changes required for comprehensive evaluation	Consider all accessibility factors and trade-offs between them. Use multimodal accessibility models.	Consider all significant impacts and planning objectives, including external costs and co-benefits.	Consider these economic efficiency factors.	Expand the range of equity impacts considered in evaluation.

This table summarizes major factors to consider in a comprehensive and multimodal congestion evaluation framework. Conventional evaluation tends to overlook and undervalue many of them.

These factors reflect different perspectives. For example, each of these perspectives can recognize benefits from improving transport options (walking, cycling, public transit, delivery services, etc.) and favoring higher value travel (HOV lanes, bus priority systems, efficient road and parking pricing). Each perspective reaches that conclusion in a different way:

- An *impact perspective* recognizes its ability to reduce problems such as traffic and parking congestion, accidents and pollution emissions.
- An *economic efficiency perspective* recognizes consumer surplus gains from serving latent demand and favoring higher value travel.
- A *social equity perspective* recognizes the value of improving transport options used by physically, economically and socially disadvantaged people.

Comprehensive evaluation incorporates different perspectives; they are not mutually exclusive.

Planning that evaluates transportation system performance based primarily on vehicle travel speed and congestion delay, and overlooks other accessibility factors and impacts, tends to exaggerate congestion compared with other transportation problems, exaggerates roadway expansion benefits, and undervalues other types of transport system improvements. These biases tend to result in more roadway capacity, reduced transport options, underpriced vehicle travel, and less accessible land use development patterns than is economically and socially optimal.

Alternatives to Roadway Level-Of-Service

This section evaluates various alternatives to roadway LOS.

Multimodal Level-of-Service (LOS) and Quality of Service (QOS)

Description: Multimodal level-of-service analysis measures travel delay experienced by pedestrians, cyclists and public transport passengers due to wider roads, heavy traffic, inadequate crosswalks, and transit delays. This analysis also measures the potential benefits of transport system changes that reduce such delays. As previously mentioned, the latest *Highway Capacity Manual* (TRB 2010) provides guidance for multimodal LOS analysis and models are now available for automating this analysis (Dowling Associates 2010).

Multimodal quality of service (QOS) analysis can account for convenience, comfort, safety and affordability in addition to speed (FDOT 2012; Fehr & Peers 2012). Since non-drivers are particularly affected by planning decisions (a motorist can purchase a more comfortable vehicle, but pedestrian, bicycling and transit comfort depends on decisions such as the sidewalk, road and transit vehicle design and maintenance), these qualitative factors are important.

Potential Criticisms: Multimodal LOS and QOS only consider travel conditions, they do not account for other accessibility factors such as transport network connectivity and land use proximity. These indicators require new data on sidewalks, crosswalks, traffic conditions and transit services, which is costly to collect.

Implementation strategies: These models already exist and can be improved with targeted research. Data collection costs can be minimized if jurisdictions establish strategic plans which begin collecting the needed data during regular field work (for example, during regular land, road and utility line surveys).

Trip Generation, Vehicle Travel and Fuel Consumption Models

Description: Trip generation models are widely used for traffic planning and are a key input into roadway LOS analysis, energy and emission modeling, traffic and environmental impact analysis.

Potential criticisms: Trip generation, vehicle travel, fuel consumption, and roadway LOS impact models are all subject to uncertainties. This uncertainty is escalated when evaluating the impacts of innovative transportation and land use changes for which there is limited experience. These changes include qualitative improvements in space-efficient modes, pricing reforms, transit-oriented development, and commute trip reduction programs (Arrington and Sloop 2010; SPACK Consulting 2010). Expanding and improving these models will require investments in research and data collection. Another possible criticism is that vehicle travel reduction targets could contradict other planning objectives, for example, by imposing restrictions that harm consumers and businesses, or by limiting development.

Implementation strategies: These models already exist and can be improved, particularly with research which identifies how various transportation demand management and Smart Growth strategies affect travel activity, and how these affect other planning objectives such as infrastructure costs, affordability, safety and health, and residents' satisfaction.

Average Travel Speeds

Most travellers are concerned with their overall travel speeds, not just congestion. Various indicators can be used to measure overall speeds. For example, the U.S. Census collects commute duration data (average minutes per commute trips), which is presented in colored heat maps in the *Commute Duration Dashboard* (<https://transweb.sjsu.edu/research/2064-Commute-Duration-Dashboard-Guide>) for all travel modes.

The study, *The Fast, the Slow, and the Congested: Urban Transportation in Rich and Poor Countries* (Akbar, et al. 2023) measured vehicle travel speed in over 1,200 large cities in 152 countries, and estimated city-level indices of travel speed and congestion. Most of the variation in urban travel speed is across countries, not within. It found urban travel is roughly 50% faster in wealthy than in poor countries, mainly because their cities have more major roads and wider land areas, which increased uncongested speed but did not significantly reduce congestion.

Potential criticisms: Many travel time data sets overlook or undercount non-auto modes. They do not reflect non-speed factors such as convenience, comfort and affordability.

Multimodal Accessibility Modeling

Description: New models evaluate accessibility based on the number of services (shops, schools, parks, etc.) and activities (such as jobs) that can be reached within a given time period and financial cost by various travel modes (Levine, et al 2012; Levinson and King 2020). Simplified versions include [WalkScore](#), [BikeScore](#), [TransitScore](#), [Transit Connectivity Index](#), [Transit Access Shed Indicator](#) and [Google Maps Commute Travel Time](#) (HTAI 2013); although these tools only reflect single modes, they can be aggregated for multimodal accessibility.

Potential Criticisms: Multimodal accessibility models are a new approach to transport system performance evaluation. They require new data and most only consider a limited set of accessibility factors. It is important that people who apply these models and their results understand their limitations.

Implementation strategies: These models are developing rapidly; they are already suitable for many planning applications (for example, even relatively crude methods such as WalkScore and Google Maps commute time applications are widely used by consumers, businesses and researchers to quantify accessibility). The availability and utility of these models are increasing rapidly. It should be possible to standardize these methods so they can be used in transport system performance evaluation.

Table 16 compares the scope of accessibility factors and impacts considered by these various evaluation methods. Roadway LOS (white square) considers just one impact for one mode: peak-period travel delay. It may measure fuel consumption and pollution emission rates per vehicle-mile, but because it does not account for per capita mileage it cannot measure total per capita fuel consumption or pollution emissions. Multimodal LOS (light blue) also considers delay to active (walking and cycling) and public transport modes. Models of vehicle trip generation, travel and fuel consumption (medium blue) can reflect additional impacts. These impacts include the costs associated with fuel consumption, emissions, parking and accidents. Multimodal accessibility models (darkest blue) also consider the effects of roadway connectivity and land use proximity on the time and other costs required to reach various destinations, and therefore accounts for the largest range of impacts.

Table 16 **Scope of Accessibility Factors and Impacts Considered**

		Accessibility Factors →						
		Automobile Travel	Active Transport	Public Transport	Roadway Connectivity	Land Use Proximity		
Impacts ↓	Traffic delay	Roadway LOS	Multimodal LOS					
	User financial costs	Vehicle Trip, Travel and Fuel Consumption Models						
	Energy consumption							
	Pollution emissions							
	Traffic safety							
	Accessibility for non-drivers	Multimodal Accessibility Models						
	Physical fitness and health							
	Land use impacts							

Roadway LOS (white square) only considers one impact (delay) for one mode (automobile). Multimodal LOS (light blue) considers delay for additional modes. Vehicle trip, travel and fuel consumption models (medium blue) indicate additional impacts. Multimodal accessibility models consider the widest range of accessibility factors and impacts, and so are the most comprehensive and multimodal.

Measuring Efficiency

Efficiency refers to the ratio of inputs (costs) to outputs (benefits). Roadway traffic efficiency can be measured in various ways that give different conclusions about what congestion reduction strategies are most efficient and beneficial overall.

- *Vehicle Travel.* Vehicle travel measures roadway efficiency based on vehicle travel speeds. This is the perspective reflected in conventional roadway performance indicators such as roadway level-of-service, traffic speeds and vehicle congestion delay.
- *Mobility.* Mobility-oriented evaluation measures roadway efficiency based on people and freight travel speeds and costs. This method recognizes that travel time savings to multi-occupant vehicles provide more benefits, and therefore more efficiency gain when compared to the same travel time savings to lower-occupancy vehicles; for example, each minute of travel time savings for a bus carrying 50 passengers has the same value as one minute saved by 50 vehicles. This perspective is multimodal; it recognizes that a portion of travelers cannot or should not drive, so a transport system is inefficient if it fails to serve these demands and forces motorists to chauffeur non-drivers. These are factors that are generally ignored with a vehicle-oriented perspective, this perspective is reflected in transport models which measure travel speeds and hours of delay per person.
- *Accessibility.* Accessibility measures transport system efficiency based on the generalized cost (time and money) required for people to access desired services and activities, and for freight to be delivered. This is the perspective reflected in transport models which measure the time or generalized costs required to access important services and activities, such as the number of jobs or commonly-used services (stores, schools, healthcare facilities, etc.) that can be accessed by residents in an area.
- *Economic Efficiency.* Economic efficiency measures roadway efficiency based on users' willingness-to-pay (wtp) for travel time savings. This method recognizes that travel time values are heterogeneous (varied); multi-occupant vehicles, commercial vehicles and travelers with urgent errands often have much higher than average time values, while some vehicle trips have only marginal net value. Users performing a trip of marginal net value would shift time, route, mode or destination if their costs increased by small amounts. Utilizing regulations or pricing to favor higher value trips and more space-efficient modes on congested roadways provides efficiency gains to transport systems. Roadway expansions may be economically inefficient if marginal costs (total roadway expansion costs plus any external costs) are less than marginal benefits (the value to users of the additional peak-period vehicle travel). This perspective is reflected in more sophisticated transport models which recognize travel time heterogeneity and calculate the net benefits gained by favoring higher value trips.

The following pages have examples of how different ways of measuring roadway efficiency can result in different conclusions about which congestion reduction strategies are best overall.

Bus Lanes

A vehicle-travel perspective evaluates transport system performance based on vehicle traffic speeds – alternative modes are only valued if they reduce automobile traffic congestion – so bus lanes are only justified if they reduce congestion delay on adjacent lanes. For example, consider an urban arterial with six lanes that each carry 800 vehicles per peak hour, including 2,250 automobiles with 1.1 average occupants and 50 buses with 40 average passengers (a bus has three passenger-car equivalents), or 2,475 automobile occupants and 2,000 bus occupants. If

evaluated using vehicle-travel indicators, a bus lane is only justified if it would cause more than a third of drivers to shift to bus travel, so the reduction in automobile capacity is more than offset by reduced automobile demand. This is a significant burden, so few arterials would have bus lanes.

A mobility-oriented perspective evaluates transport system performance based on person-speeds, using models that measure total traveler time costs. This perspective recognizes that a minute saved by a bus carrying 40 passengers is worth about 36 times as much as a minute saved by an automobile carrying 1.1 occupants. These numbers justify bus lanes even if they slightly reduce automobile travel speeds, provided the increased automobile occupant travel times are more than offset by the total travel time savings from bus passengers. Bus lanes are also justified on most urban arterials that have more than about 24 buses during peak hours since those buses carry more passengers than a general traffic lane. Mobility-oriented indicators recognize that a road system becomes more efficient if it favors space-efficient modes over space intensive modes (an automobile traveler requires 10 to 100 times as much road space as a bus passenger).

An accessibility-oriented perspective recognizes that travel times and costs should be measured door-to-door, rather than on individual links. Measuring this way recognizes the efficiency gains that result from more integrated transport networks (more connected road networks and better connections between walking, cycling, automobile and public transit services) as well as more accessible land use development. Accessibility-oriented evaluation supports bus lanes integrated with transit-oriented development, for example, having bus lanes that connect major employment, education, shopping, healthcare and recreation centers.

An economic efficiency perspective recognizes all of the previously described factors (the value of favoring higher-occupancy vehicles). Utilizing economic efficiency also recognizes that some vehicle trips have higher economic value than others, and that some peak-period vehicle trips may have marginal net value, and so are quite price sensitive. This perspective justifies efficient pricing that gives higher-value trips priority over lower-value trips and tests users' willingness-to-pay for road improvements; this prevents society from spending \$2.00 to provide additional road capacity that users only value at \$1.00; such capacity expansion would be economically inefficient. Efficient pricing favors more space-efficient travel (bus occupants pay less per passenger-mile than car occupants) and higher value trips (vehicle users can pay for faster travel for commercial vehicles and urgent errands). This method can avoid the need for special bus and truck lanes.

Lower Speed Urban Roadways

In their study, *Tradeoffs Among Free-flow Speed, Capacity, Cost, and Environmental Footprint in Highway Design*, Ng and Small (2012), conclude that urban arterials with narrower traffic lanes (e.g., 10-feet) and lower design speeds (e.g., 45 miles-per-hour) often carry more capacity than higher speed urban highways. These arterials also have lower construction costs and environmental impacts as well as better aesthetics. Ng and Small (2012) conclude that more comprehensive analysis of these tradeoffs would probably result in fewer urban highways and more medium-speed urban arterials when compared to what results from current planning that emphasizes the value of traffic speed and has dedicated highway funding.

Active Mode (Pedestrian and Cycling) Improvements

Active mode improvements can include enhanced sidewalks, paths, crosswalks and bicycle parking facilities, reduced vehicle traffic speeds and the use of other traffic management strategies that make active modes more convenient and attractive to use.

A *vehicle-travel perspective* assumes that the primary planning goal is to maximize travel speed, and so tends to consider walking and cycling inefficient and of little value. For example, most traffic modeling recognizes the delay that increased traffic imposes on other motor vehicles but ignores the delay it imposes on active modes (called the *barrier effect*). *Mobility-, accessibility- and economic-efficiency-oriented perspectives* tend to recognize the various roles that walking and cycling play in an efficient transport system. The roles of these active modes include mobility to non-drivers (and therefore reducing the need for drivers to chauffeur non-drivers), public transit access, and support for more compact development.

Pricing Reforms

Efficient transport pricing includes road tolls, parking fees, distance-based insurance premiums and fuel tax increases that charge motorists more directly for the costs imposed by their vehicle use (currently only about half of roadway costs and an even smaller portion of non-residential parking costs). A *vehicle-travel perspective* assumes that increasing vehicle travel and vehicle travel speeds are inherently beneficial and that demand management strategies are solutions of last resort to be applied only where roadway expansion is infeasible. It tends to consider road tolls as a roadway expansion finance strategy and generally opposes applying tolls on existing roadways (Poole 2009).

Mobility-, accessibility- and economic-efficiency-oriented perspectives recognize that efficient pricing favors space-efficient modes and higher-value trips, and so can increase transport system efficiency by encouraging travelers to shift the following:

- From peak to off-peak travel.
- From space intensive modes (automobiles) to more space efficient modes (walking, cycling and public transit).
- From far destinations to closer ones.

Smart Growth Policies

Smart Growth refers to policies that encourage more compact, mixed and multimodal development. A *vehicle-travel perspective* tends to consider compact development inefficient since increased density may reduce traffic speeds (Melia, Parkhurst and Barton 2011). Empirical evidence suggests that reduced travel speeds are often offset by reduced trip generation (Kuzmyak 2012).

Mobility-, accessibility- and economic-efficiency-oriented perspectives recognize that Smart Growth can increase transport system efficiency if it encourages mode shifts and shorter travel distances. For example, Smart Growth recognizes that locating a school toward the center of a residential neighborhood may be more efficient than locating it on a major roadway because more students can walk or bike to school, and travel distances are shorter. An *economic efficiency perspective* also recognizes the value of charging motorists directly for using roads and parking facilities, or the value from offering comparable benefits to employees and students who use other travel modes that reduce their road and parking facility costs.

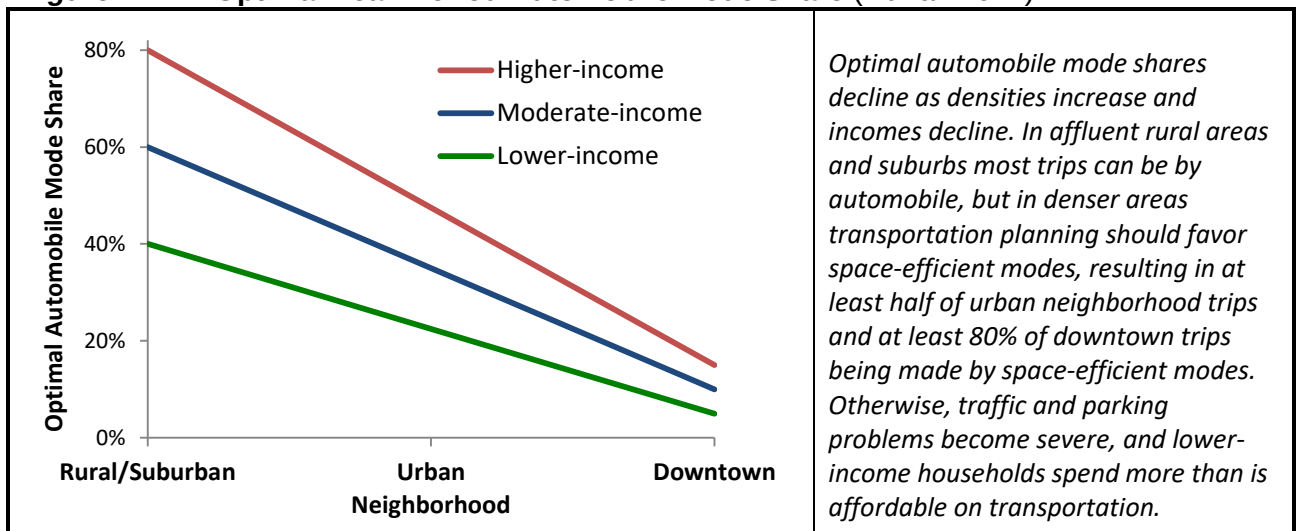
Optimizing Urban Accessibility

Cities emphasize *accessibility* by locating activities close together instead of increased mobility (travel speed). Urban residents often have more services and jobs within a five-minute *walk* than suburban and rural residents have within a five-minute *drive*. Urban residents can drive less, spend less on transport, impose lower road and parking costs, have lower crash rates, and produce less pollution than residents of automobile-oriented locations.

Proposals for infill development are often opposed on the grounds that they will increase congestion, but that is not necessarily true; it ignores how cities work. Adding more people to an area increase *trips*, but with good planning this is offset by shorter trip distances and reduced automobile mode shares. Although congestion may become more intense, per capita delays tend to decline, particularly if infill development is accompanied with improvements to space-efficient modes, TDM incentives such as efficient parking pricing and commute trip reduction programs, and with more mixed development so most common services are available within each neighborhood.

Automobile travel requires more road space, and so imposes more congestion costs than other modes. This type of travel also imposes more pedestrian delay, accident risk and parking and pollution costs per passenger-mile compared to other modes. As a result, transport system efficiency, economic productivity, affordability and community livability tend to increase if urban automobile travel is minimized, particularly under urban-peak conditions. Automobile travel is not eliminated, but as cities become larger and denser, or residents become financially burdened vehicle costs, automobile mode share should decline, illustrated in Figure 14.

Figure 14 Optimal Peak-Period Automobile Mode Share (Litman 2017)



Current Congestion Evaluation

Various studies evaluate congestion costs and potential congestion reduction benefits, including targeted studies such as the *Urban Mobility Report* (TTI 2012), the USDOT's *Conditions and Performance Report to Congress* (USDOT 2010), and various benefit/cost models used to value transport programs and projects. Table 17 evaluates the degree that each study considers in regard to the factors required for comprehensive congestion evaluation.

Table 17 Evaluating the Scope of Current Congestion Cost Studies

Studies	Accessibility	Impacts	Economic Efficiency	Social Equity
	- Automobile travel quality - Quality of other modes - Roadway connectivity - Geographic proximity (land use density and mix)	- Traffic congestion - Road & parking costs - Accidents - Consumer costs - Mobility for nondrivers - Energy consumption - Pollution emission - Efficient land use - Public fitness	- Efficiency gains from favoring higher value trips - Possibility that induced travel has negative net benefit - Consumer surplus gains from new modes and services	- Fairness of benefit and costs allocation. - Impacts on physically, economically and socially disadvantaged people. - Unfairness of policies that favor automobile travel over other modes
Conditions and Performance Report (annual report to Congress on transport system quality)	Considers highway and transit conditions, and discusses walking and cycling. Measures congestion using the travel time index. Does not account for roadway connectivity or land use accessibility.	Considers congestion, accidents, energy consumption, and pollution emissions, plus livability which could account for other factors such as public fitness and affordability.	Discusses congestion pricing and other pricing reforms, and includes analysis of the degree that roadway expenditures are covered by user fees.	Primarily concerned with geographic equity (whether benefits are distributed fairly between jurisdictions). Some discussion of basic mobility (portion of residents who have transit service available).
Urban Mobility Report (widely cited study of congestion costs and potential congestion reduction strategies)	Although it includes various congestion indicators, comparative analysis is based on the travel time index. Walking, cycling and transit are only considered if they affect automobile congestion.	Measures travel time, vehicle operating costs and pollution emissions. Ignores induced travel impacts.	Mentions road pricing as a possible congestion reduction strategy, but does not mention any other economic efficiency issues.	Ignores equity impacts. Tends to assume that transportation means driving (the terms "commuter" is often used when the analysis only considers automobile commuters while users of other modes are ignored).
Highway Capacity Manual (widely used roadway engineering manual)	The 2010 version includes level-of-service indicators for walking, cycling, and public transit plus automobile conditions.	Primarily measures travel time and safety, but implicitly considers basic mobility for non-drivers.	Does not explicitly evaluate economic efficiency or consumer surplus impacts.	Does not explicitly evaluate social equity impacts but does support improved mobility for non-drivers.
Benefit/cost models used to evaluate specific projects	Generally use network models to measure congestion delays, which often measure multiple modes and land use factors.	Primarily measures travel time, vehicle operating costs, crash rates and pollution emissions. Usually ignores other impacts.	Does not usually evaluate economic efficiency or consumer surplus impacts.	Does not usually evaluate social equity impacts

This table evaluates the degree that various transportation studies account for the various factors required for comprehensive and multimodal evaluation. Current studies do not account for many of these factors.

Most congestion evaluation studies overlook important factors required for comprehensive and multimodal evaluation (Bain 2009; Cortright 2011; Litman 2019). These studies evaluate congestion intensity rather than total congestion costs; they ignore common trade-offs between accessibility factors, such as when roadway expansion creates barriers to walking and bicycling, hierarchical road networks reduce roadway connectivity, or stimulates sprawled development. Most ignore induced travel effects (Volker, Lee and Handy 2020), which exaggerates roadway expansion benefits and overlooks the additional external costs. Such studies also tend to overlook co-benefits provided by improvements to non-auto modes and transportation demand management strategies. For example, they often ignore the vehicle ownership and parking savings from improvements to non-auto modes. Because of this, the savings to households and businesses that result in improved commute options reduce automobile trips and allow some households to reduce their vehicle ownership.

The *Urban Mobility Report's* title implies that it evaluates overall urban transportation performance, but it actually only considers motor vehicle traffic congestion; it includes no analysis of other urban transport issues such as walking and cycling conditions, public transit service quality, parking congestion, affordability, accident risk, or overall energy consumption and pollution emissions. For accuracy, it should be renamed the *Urban Traffic Congestion Report*.

Few congestion evaluation studies reflect economic principles. Most studies do not account for the economic efficiency gains provided by road pricing or other strategies that favor higher-value trips. Nor do these studies factor in the possibility that roadway expansions could be inefficient if the marginal costs (roadway expansion costs plus any external costs of induced travel) are worth less than marginal benefits; an exception is the *Conditions and Performance Report*, which discusses these issues.

The scope of transport project economic evaluations varies depending on the quality of input data, the type of transport modeling (particularly the sensitivity of the model to factors such as transit service quality and congestion feedback, and whether it can report induced travel impacts), and the range of impacts considered. For example, transport models can predict how projects and programs affect automobile ownership and trip generation and therefore, impacts on vehicle ownership and parking cost. If models accurately measure how transport system changes affect walking, cycling, public transit and automobile access, they can disaggregate impact by user type. It is also possible to estimate the benefits for economic efficiency that are provided by policies that favor higher-value vehicle travel, and the consumer surplus impacts of price changes or serving latent demands. It is therefore possible for transport models to provide much more comprehensive and multimodal analysis of how potential congestion reduction strategies affect overall transport system performance.

Economic Productivity Impacts

Motor vehicle travel is critical for many economic activities, including deliveries, commuting, shopping and tourist activities. Congestion can reduce productivity by delaying these economic activities (EDRG 2007). However, as previously discussed, vehicle travel speeds are only one factor in overall accessibility; for example, a business can generally access more goods, employees and customers if located in a dense, congested urban area rather than in a sprawled, less congested area (Levine, et al 2012; Levinson and King 2020; RPA 2014). Businesses use various techniques to minimize their congestion costs, for example, by shipping goods during off-peak periods and using traffic information services such as TomTom and INRIX to avoid congestion. Regional productivity tends to increase with congestion and declines with increased vehicle travel and road supply, as illustrated in figures 15 to 16.

Figure 15 U.S. Metro Region Traffic Delay and GDP (Dumbough 2012)

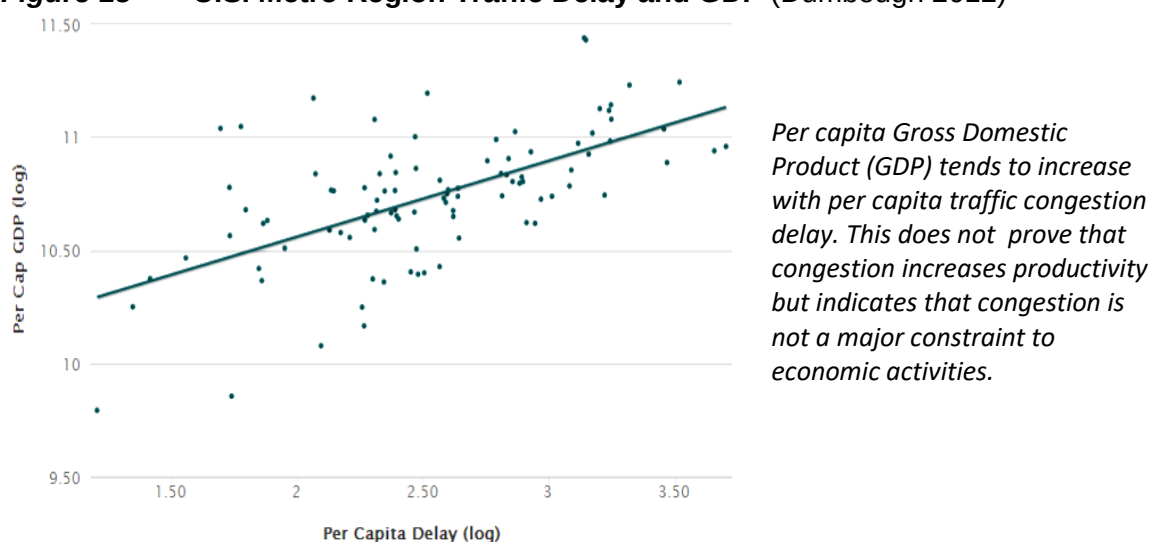
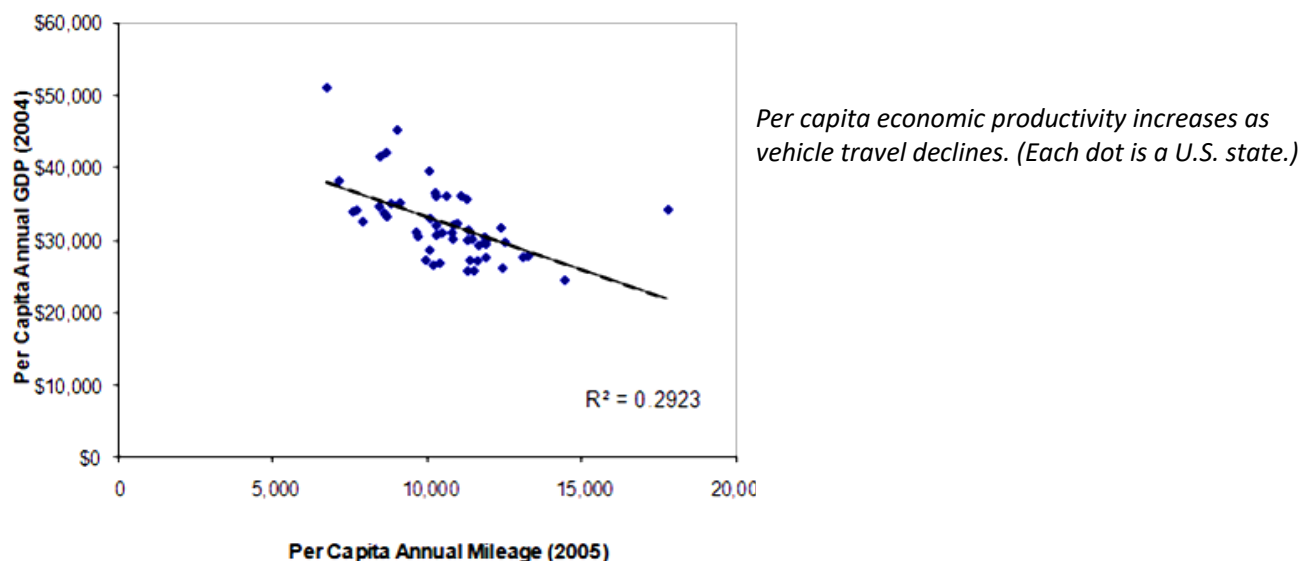


Figure 16 Per Capita GDP and VMT for U.S. States (VTPI 2009)

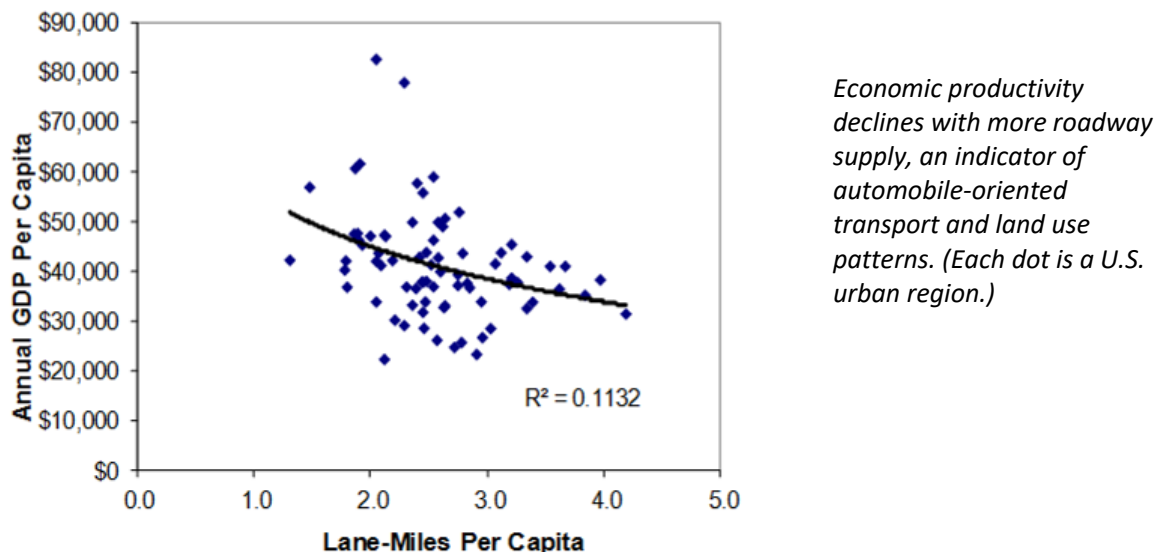


Analyzing the effects of roadway investments on economic output and induced travel demand in U.S. urban regions, Melo, Graham and Canavan (2012) found that roadway expansions are associated with increased average economic growth, but induce additional motor vehicle travel, resulting in no long-term reductions in traffic congestion. They conclude that transportation demand management strategies are likely to be more effective than roadway expansions at improving roadway system performance and supporting economic development.

Marshall and Dumbaugh (2018) used sophisticated statistical analysis of 30 years of data for 89 US metropolitan statistical areas (MSAs) to evaluate traffic congestion's regional economic impacts. Controlling for the key variables, they found a *positive* association between traffic congestion and per capita GDP as well as between traffic congestion and job growth. In addition, congestion impacts on per capita income, although negative, is statistically insignificant. They conclude that "There may be valid reasons to continue the fight against congestion, but the idea that congestion will stifle the economy does not appear to be one of them."

This evaluation does not prove that traffic congestion is economically beneficial, but suggests that at worst it is a minor constraint on productivity and its negative impacts are overwhelmed by other accessibility and cost factors (Sweet 2013). For example, the *Urban Mobility Report* estimates that traffic congestion increases trucking costs by \$27 billion annually, or about 5% of the industry's total costs (TTI 2012). The trucking industry generally opposes decongestion pricing despite its potential effectiveness at reducing their delays, suggesting that the industry considers congestion a modest problem (Boyce 2009).

Figure 17 Per Capita GDP and Road Lane Miles (VTPI 2009)



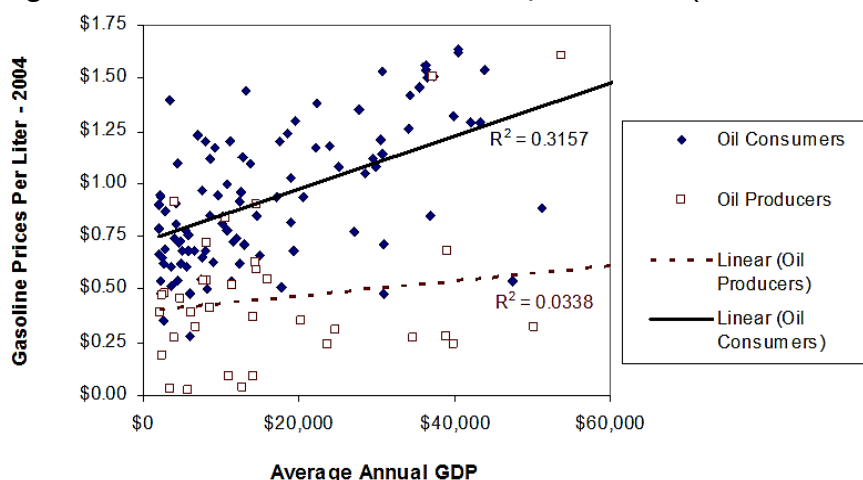
Congestion reduction strategies vary in their economic productivity impacts:

- Urban roadway expansions have mixed economic impacts. Although many economic activities depend on road transport, once a basic road network exists, there is little evidence supporting that expanding its capacity increases productivity (Nadri and Mamuneas 1996 and 2006). Urban roadway expansions tend to reduce congestion in the short-run, but as previously discussed, this benefit tends to decline over time as generated traffic fills the

additional capacity. Most of the additional vehicle travel is personal travel, for example, additional capacity gives households more dispersed housing and shopping options; there is often little savings to commercial travelers. Roadway expansions can increase congestion and other transport costs over the long run by reducing other forms of access, for example, by creating barriers to walking or stimulating sprawled development.

- Improving space-efficient modes, particularly grade-separated public transit, tends to reduce peak-period vehicle travel which reduces traffic congestion costs, expands labor pools (including for non-drivers), reduces parking costs and vehicle/fuel expenditures, and tends to stimulate more accessible development.
- Transportation pricing reforms (efficient road and parking fees, fuel tax increases and distance-based insurance) tend to reduce total vehicle travel and associated external costs, including congestion, facility costs, traffic accidents, fuel consumption and pollution emissions. Their impacts vary depending on the type of pricing and specific conditions. For example, decongestion pricing (road tolls with higher fees during congested periods) is particularly effective at reducing congestion, while fuel tax increases are particularly effective at reducing the economic costs of importing and consuming vehicle fuel. Economic theory suggests that to the degree that vehicle travel imposes external costs, pricing reforms should increase economic efficiency and productivity. Available evidence indicates that all else being equal, higher vehicle user fees are associated with increased per capita economic productivity, as indicated in Figure 18.

Figure 18 GDP Versus Fuel Prices, Countries (Litman 2014c)



Economic productivity tends to increase with higher fuel prices, indicating that substantial increases in vehicle fees can be achieved without reducing overall economic productivity.

- Smart Growth development policies may increase congestion intensity (Steve, Parkhurst and Barton 2011), but by reducing travel distances and improving mobility options, tends to reduce per capita congestion costs (Kuzmyak 2012).
- Transportation demand management (TDM) programs, such as commute trip reduction programs and mobility management marketing tend to reduce peak-period vehicle travel which reduces traffic congestion and often improves alternative modes. Improving alternative modes tends to expand labor pools (including non-driver), reduces parking costs and vehicle/fuel expenditures, and tends to support more accessible development.

As discussed earlier in this report, traffic congestion costs are modest compared with other transportation costs. A congestion reduction strategy provides smaller net benefits and less productivity gains if it increases other economic costs. These other costs include road and parking infrastructure, accident and pollution damages, or the economic costs of importing vehicles and fuel. A congestion reduction strategy provides far greater benefits if it reduces these other costs.

Table 18 summarizes the economic development impacts of various congestion reduction strategies. Of course, these impacts can vary significantly depending on specific factors. In some situations, urban roadway expansions may support economic development, although, this support is often less than other congestion reduction strategies. These alternative strategies tend to have synergistic effects – their total impacts are greater than the sum of their individual impacts. Because of the large scale of the total impacts, these strategies should generally be evaluated and implemented as an integrated program that includes appropriate improvements to space-efficient modes, pricing reforms, Smart Growth development policies and TDM programs.

The above evidence and other research suggest that urban roadway expansions provide less net benefits than other congestion reduction strategies such as improving space-efficient modes, more efficient transport pricing, and other TDM strategies (Cambridge Systematics 2012; Jiwattanakulpaisarn, Noland and Graham 2012; Metz 2021).

Table 18 Economic Impacts of Congestion Reduction Strategies

Economic Impacts	Roadway Expansion	Improve Alt. Modes	Efficient Pricing	Smart Growth	TDM Programs
Traffic congestion	Reduces short-run intensity, but increases long-run costs	Reduces congestion	Reduces congestion	Increases intensity, reduces total costs	Reduces congestion
Labor pools	Expands car commuters' work options	Expands all commuters' work options	Expands most commuters' work options	Improves worker accessibility	Can improve access
Parking costs	Increases parking costs	Reduces parking costs	Reduces parking costs	Increases unit costs but reduces total costs	Reduces parking costs
Vehicle and fuel imports	Increases	Reduces	Reduces	Reduces	Reduces
Land use accessibility	Causes sprawl, which reduces accessibility	Encourages compact development which improves accessibility	Encourages compact development which improves accessibility	Increases land use accessibility	Supports more accessible development

Roadway expansions can reduce congestion in the short-run, but do little to improve non-drivers' work options, and can have undesirable economic impacts including increased parking costs, vehicle and fuel imports, and sprawl. Other congestion reduction strategies often provide more economic benefits.

Evaluating Potential Congestion Reduction Strategies

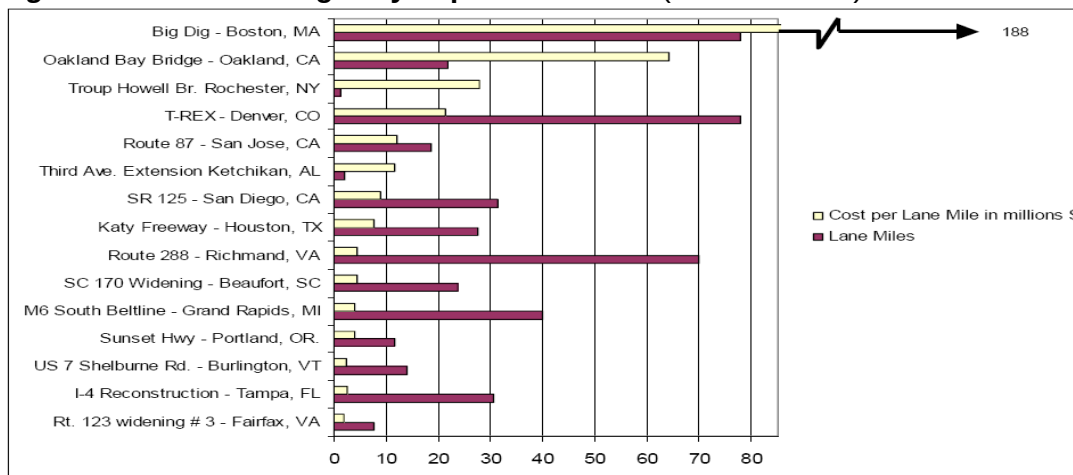
This section describes and compares various congestion reduction strategies. Also see ITF (2021).

Roadway Capacity Expansion

Roadway capacity expansions can include new and expanded roads and bridges, wider and straighter lanes, intersection flyovers, traffic signal synchronization, reduced cross-streets and crosswalks on arterials, reversible lanes, conversions from two-way to one-way streets, automated highway technologies, half-width vehicles, improved incident response, and various transportation systems management (TSM) strategies. Automobile-oriented planning considers these the preferable solutions to traffic congestion (AHUA 2004).

Although some roadway expansion strategies, such as signal synchronization, are relatively inexpensive, most are costly (“Roadway Costs,” VTPI 2012). Urban road expansions often cost \$10-20 million per lane-mile, including land, pavement, and intersection reconstruction expenses, as illustrated in Figure 20. This represents an annualized cost of \$300,000-700,000 per lane-mile (assuming a 7% interest rate over 20 years). Dividing this by 4,000 to 8,000 additional peak-period vehicles for 250 annual commute days indicates costs of 15¢ to \$1.00 per additional peak period vehicle-mile, and sometimes more.

Figure 20 Urban Highway Expansion Costs (WSDOT 2004)



Of 36 highway projects studied by the Washington State Department of Transportation, 13 had costs exceeding \$10 million per lane-mile. Future projects are likely to have higher unit costs since most jurisdictions have already implemented the cheapest highway projects.

Roadway planning decisions often face a paradox: motorists demand costly roadway expansions provided somebody else bears the costs, but if charged marginal cost pricing, such as road tolls that are high enough to repay the project costs, demand declines significantly. Such projects are therefore economically inefficient and unfair (Hau 1998). Tolls of 20-30¢ per vehicle-mile typically reduce traffic volumes by 20-30%, and more if there are good alternative routes and modes (Spears, Boarnet and Handy 2010). Many recent toll road projects have failed to achieve their traffic volumes and revenue targets (NCHRP 2006; Prozzi, et al. 2009). As a result, user fee revenue is seldom sufficient to fully finance urban roadway expansions. This data indicates that roadway expansion is seldom economically efficient: users only want the additional capacity if it is subsidized.

There is debate concerning how much urban roadway expansion reduces congestion. Roadway expansion usually provides only modest and short-term congestion reductions on major urban corridors where congestion is most intense (Litman 2001; Handy and Boarnet 2014). The *Urban Mobility Report* claims that highway expansions reduce congestion growth rates, as illustrated in Figure 21, but their analysis fails to account for differences in city size and growth rates that affect congestion growth. The reports also measures congestion intensity instead of total congestion costs and so does not account for increased total delays caused by sprawl.

Figure 21 Congestion Growth Versus Highway Expansion (TTI 2012, p. 20)

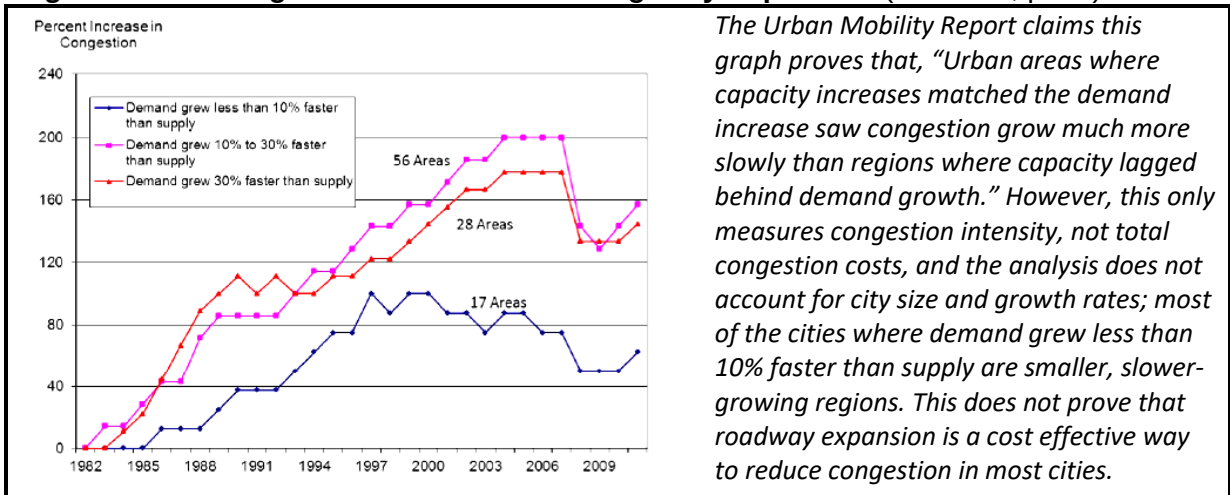
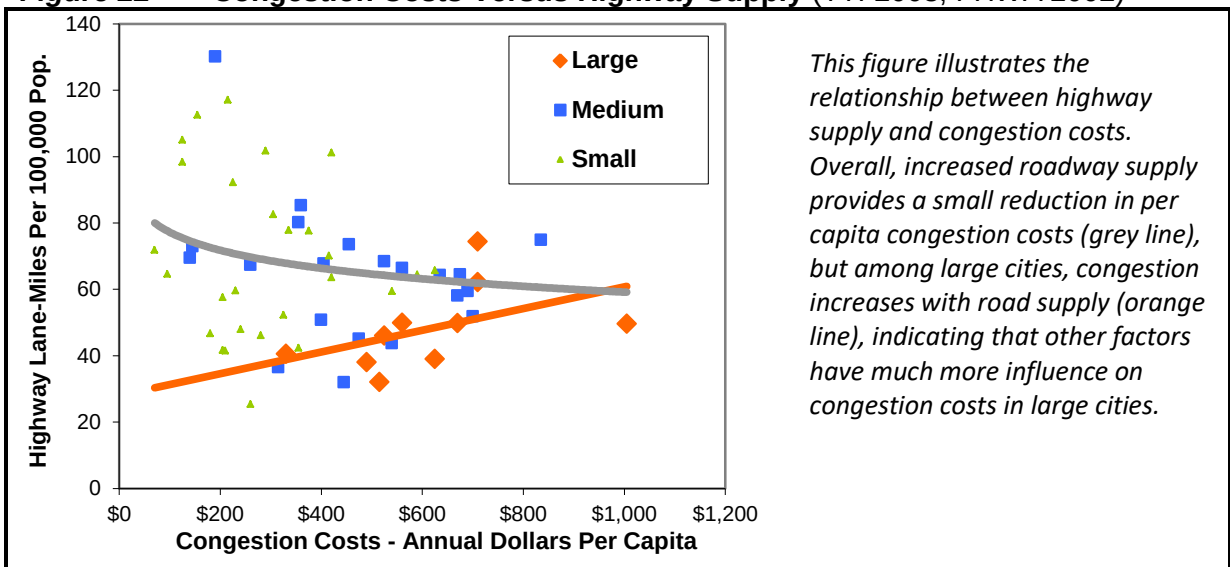


Figure 22 illustrates the relationship between urban highway lane-miles and congestion costs. Considering all cities, congestion decreases with more lane-miles, but the relationship is weak (grey line). Among the ten largest cities (orange diamonds), the relationship is negative (orange line): those with more highways tend to have higher per capita congestion costs, probably because increased highway capacity increases automobile dependency and sprawl.

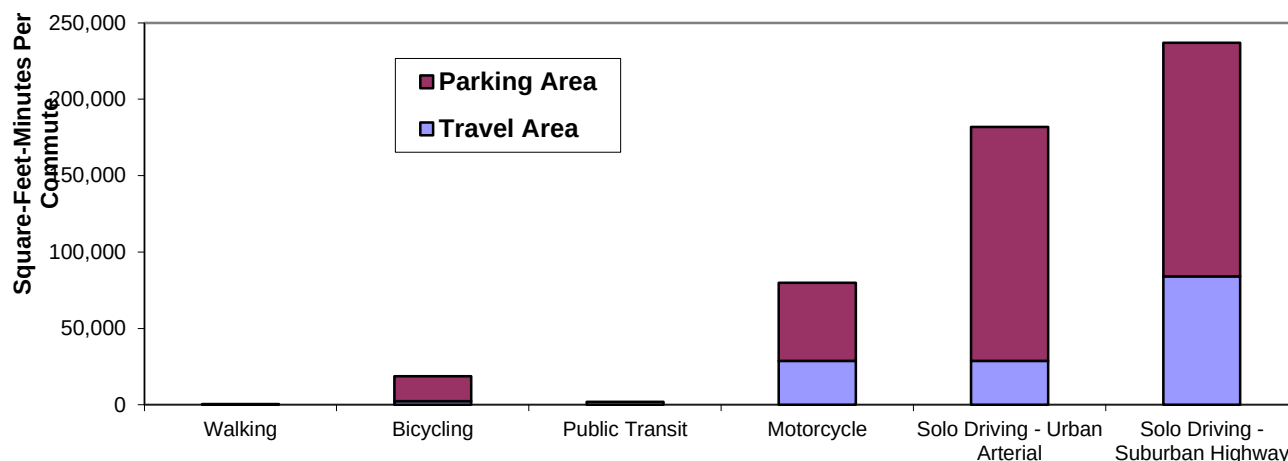
Figure 22 Congestion Costs Versus Highway Supply (TTI 2003; FHWA 2002)



Improving and Favoring Space-Efficient Modes

The amount of space required to travel tends to increase with vehicle size and speed. For example, a car traveling at 30 miles-per-hour (mph) requires about 12 feet of lane width and 60 feet of lane length, or about 720 square feet in total, but at 60 mph this increases to 15 feet of lane width and 140 feet of length, or about 2,100 square feet. A bus requires about three times as much road space (measured as “passenger car equivalents”), but typically carries 30-60 times as many passengers under urban-peak conditions. Vehicles also require space for parking at each destination. Figure 23 compares the travel and parking space required for commuting by various modes.

Figure 23 Space Required by Travel Mode (www.vtpi.org/Transport_Land.xls)



Walking, cycling and public transit are space-efficient compared with automobile travel.

If space-efficient modes are inconvenient, uncomfortable, dangerous, or unaffordable, travelers will drive even if congestion is severe. As space-efficient modes’ service quality improves, travelers’ are more likely to shift mode, reducing the point of congestion equilibrium. Even small shifts can provide significant benefits. For example, a 5% reduction from 2,000 to 1,900 vehicles per lane-hour typically increases traffic speeds by 5-15 miles per hour.

Table 19 Typical Alternative Mode Improvements

Walking	Bicycling	Public Transport
• More sidewalks and paths • More crosswalks • Traffic speed reductions • Improved wayfinding • More compact and mixed development so more services are within walking distance • Improved safety and security • <i>Universal design</i>, so pedestrian facilities accommodate pedestrians with disabilities • Improved connectivity	• More paths • More bike lanes • Traffic speed reductions • Improved wayfinding • Bike parking • Bike racks on buses and trains • Improved safety and security • Bicycle training and encouragement programs • Loans and subsidies to purchase bicycles	• More routes • More frequent service • Faster service, grade separation • Higher quality vehicles and stations • Improved connections • Improved user information • Improved safety and security • Reduced fares • More convenient payment systems • Improved stop/station access • Better marketing • <i>Universal design</i>

There are many possible ways to improve space-efficient modes.

How Improving Transport Options Can Reduce Traffic Congestion

Urban traffic congestion tends to maintain equilibrium, it grows to the point that congestion delays discourage additional peak-period vehicle trips. If congestion increases, some travelers change route, destination, travel time and mode to avoid delay, and if it declines they take more peak-period trips. Reducing the point of equilibrium is the only way to achieve durable congestion reductions.

The quality of travel options influences the point of congestion equilibrium: If alternatives are inferior, fewer motorists will shift mode and the point of equilibrium will be high. If alternatives are attractive, travellers are more likely to shift modes, reducing the point of equilibrium. To attract discretionary riders (travelers who have the option of driving), transit must be fast, comfortable, convenient and affordable. Grade-separated service (such as rail on separate right-of-way or busways) provides a speed advantage that can attract discretionary riders. When transit is faster than driving, a portion of travelers shift mode until the highway reaches a new equilibrium (that is, until congestion declines to the point that transit is no longer faster). Several studies find that door-to-door travel times for motorists tend to converge with those of grade-separated transit (Vuchic 1999). The actual number of motorists who shift to transit may be relatively small, but is enough to reduce delays. Congestion does not disappear, but it never gets as bad as would occur if grade-separated transit service did not exist nearby. As a result, improving travel options can increase travel speeds for both travelers who shift modes and those who continue to drive.

The article, *Traffic Congestion Relief Associated with Public Transport: State-of-the-Art* (Nguyen-Phuoc, et al. 2020) evaluates the degree that various congestion indicators reflect public transit's congestion reduction impacts. The study concludes that public transit does reduce congestion, but because most indicators focus on vehicles rather than people, they tend to underestimate these impacts.

Sophisticated analysis of 545 European cities indicates that urban highway expansion tends to increase vehicle traffic and so fails to solve congestion (Garcia-López, Pasidis, and Viladecans-Marsal 2020). The study indicates that each 1% increase in highway lane-kilometers typically increases total vehicle kilometers by 1.2%. The analysis found significantly less congestion (indicated by vehicle-kms relative to the log of lane-kms) in cities with road pricing and high quality rail transit. A 1% increase in lane kilometers increases congestion by 1.9% in cities without highway tolls but only 0.3% in cities with tolls. A 1% increase in railroad network length decreases congestion by 0.6% in a city without subways, 0.8% in a city with the average share of subways, and 1.3% in a city where the majority of the railroad network consists of subways.

Recent studies indicate that by increasing urban automobile traffic ridehailing services such as Uber and Lyft increase congestion (Fageda 2021). For example, Diao, Kong and Zhao (2021) found that the entrance of ridehailing services in a community increased road congestion intensity (by 0.9%) and duration (by 4.5%), and transit ridership by 8.9%. Ridehailing services increase the need for urban traffic management, as discussed in *Shared Mobility Principles for Livable Communities* (www.sharedmobilityprinciples.org).

Various policies and technologies can favor space-efficient modes. These include bike, HOV and bus lanes (Litman 2015a), bus prioritization in traffic signals and intersection designs, and road pricing.

Active Modes (Walking and Cycling)

Walking and cycling improvements can reduce traffic congestion in several ways (ITF 2021). Poor walking and cycling conditions force people to drive for even short trips. A significant portion of urban vehicle traffic (typically 10-30%) consists of short trips suitable for active modes. Poor walking and cycling conditions also force motorists to chauffeur non-drivers to local destinations; such trips often include empty backhauls, so each passenger-mile generates two vehicle-miles of travel. Since most public transport trips include walking and cycling links, improving these modes tends to increase transit travel.

For example, consider how walking and cycling improvements can reduce traffic congestion around a 500-student primary school. If improvements allow the portion of students driven by parents to decline 20-percentage points, this reduces 200 peak-period vehicle trips (including return trips), significantly reducing local traffic and parking congestion. Similarly, if walking and cycling improvements at a commercial district with 1,000 peak period customers reduce the average number of between-store vehicle trips per shopper from 4 to 3, this reduces local circulation trips from 4,000 to 3,000, which can significantly reduce parking lot and local roadway congestion.

Hamilton and Wichman (2016) use a unique fine-grained traffic dataset to measure the Washington DC Capital Bikeshare program's impacts on congestion. They find that bikeshare stations reduce traffic congestion by 4% or more compared with congestion intensity that would otherwise occur, with the greatest reductions in the most congested areas.

Non-motorized traffic can also contribute to congestion. Pedestrians primarily cause delays when crossing or walking on roads that lack sidewalks. To analyze the bicycling congestion impacts, road conditions are divided into four classes:

1. *Uncongested roads.* Bicycling in these conditions causes no congestion.
2. *Congested roads with space for bicyclists.* Bicycling on road shoulders (common rural roads), wide curb lanes (common in suburban and urban areas), or bike lanes causes little congestion except at intersections where turning vehicles may be delayed.
3. *Narrow, congested roads with low-speed traffic.* Bicycling on low-speed streets where cyclists keep up with traffic (common on urban streets) usually causes less congestion than an average car due to bicycles' smaller size.
4. *Narrow, congested roads with moderate to high-speed traffic.* Bicycling on a narrow, congested road where faster vehicles cannot easily pass can cause significant delay.

The *FLOW Multimodal Transport Analysis Methodology and Impact Assessment Tool* (www.h2020-flow.eu) evaluates active transport impacts on transport system performance. Case studies indicate that walking and bicycling improvements generally reduce congestion (Rudolph 2017).

Travelers shifting from driving to bicycling under the first three conditions reduces congestion. Only under condition 4 are shifts likely to increase congestion. Condition 4 represents a small portion of total cycling because most bicyclists avoid such conditions.

High Occupant Vehicles (HOVs)

High Occupant Vehicles (HOVs) include rideshare (car- and vanpool) vehicles and buses. Some roads have HOV lanes which may only be used by vehicles with a minimum number of occupants, which typically range from two (2+) to seven (7+). HOV lanes can carry more people than general traffic lanes, which increases roadway efficiency (more passengers per lane-hour). Their higher speeds also may attract some travelers who would otherwise drive, which can reduce traffic congestion. However, such mode shifting is usually modest, only a few percent of corridor travelers. This modest shift is due to most HOV lanes only affecting a minor portion of commuters' total trips; to be effective HOV priority lanes must be implemented with other mode shift incentives such as efficient road and parking pricing, and overall public transit service improvements (VTPI 2012).

Mode Shifting Economics

A ridesharing or public transit improvement or incentive often causes only modest mode shifting since it only affects a minor portion of total travel costs. For example, an HOV or bus lane might increase speeds by 30% on that roadway link, providing five minutes of travel time savings. However, for a typical 50-minute commute trip, that only represents a 10% savings, and assembling a rideshare or catching a bus often adds 10-20 minutes. A five-minute time savings may induce some mode shifting, but usually just a few percent of total trips.

To cause significant mode shifting and congestion reductions, HOV and bus lanes must be implemented with other service improvements and incentives, such as increased transit service, nicer vehicles and stations, amenities such as on-board Internet service (particularly for express commuter buses), financial incentives such as parking pricing or cash-out, and commute trip reduction and mobility management marketing programs (VTPI 2012). By providing a combination of incentives to shift mode these often have synergistic effects (their total impacts are greater than the sum of their individual impacts), and because ridesharing and public transit services have scale economies (unit costs decline as demand increases), such integrated policies and programs are often cost effective. Described differently, HOV and bus lanes become more cost-effective if implemented with ridership incentives, and ridership incentives become more cost effective if implemented with HOV and bus lanes.

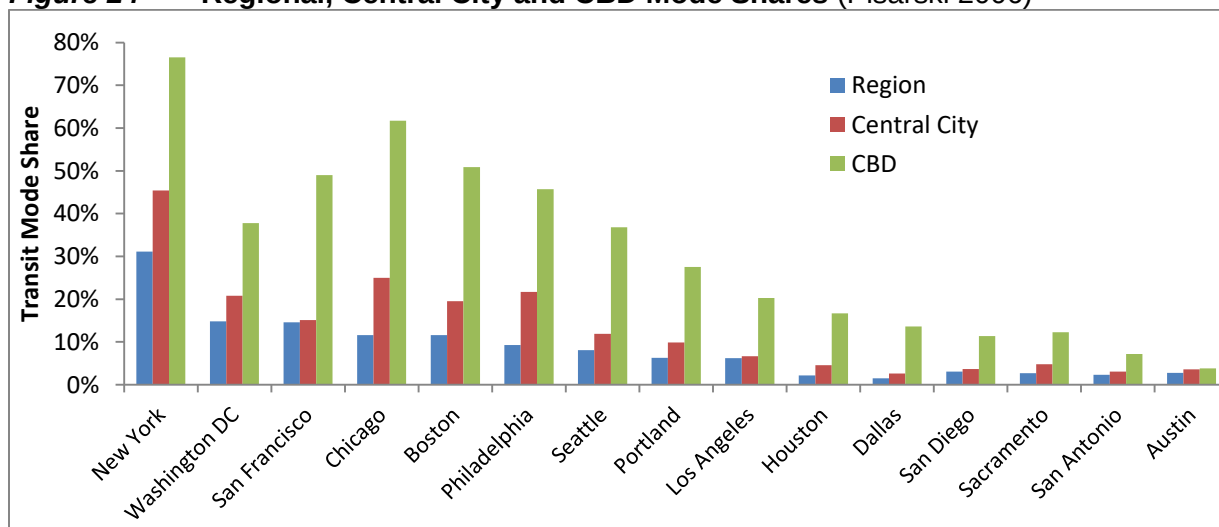
Public Transport

High-quality public transit, which attracts discretionary travelers (people who would otherwise drive) can reduce the point of congestion equilibrium. These benefits can be difficult to measure because of confounding factors: congestion and transit ridership both tend to increase with city size, density, transit service quality and employment rates (traffic congestion and transit ridership tend to increase with a business cycle). Studies that account for these factors generally indicate that public transit service improvements reduce traffic congestion intensity and costs (Nelson\Nygaard 2006; Nguyen-Phuoc, et al. 2020).

Studies indicate that peak-period highway travel times tend to converge with transit travel times within a corridor. For example, if a suburb-to-city commute takes 30 minutes by transit, traffic congestion on parallel roadways will decline to the point that automobile commutes take a similar amount of time (Vuchic 1999). As a result, grade-separated services, such as bus-lanes and trains on their own rights-of-way, are particularly effective at reducing congestion. Other factors that attract discretionary transit travelers, such as improved convenience, comfort and affordability, are also likely to reduce congestion on parallel roadways.

Public transit only carries a minor portion of total regional travel, but its mode share tends to be much higher on congested urban corridors and in central business districts (CBDs) and so can provide significant congestion reduction impacts (Figure 24). For example, although Los Angeles has only 11% transit commute mode share, one study found that transit reduces regional congestion costs by 11% to 38%. When a strike halted transit service for five weeks in Los Angeles, average highway congestion delay increased 47% (Anderson 2013), with particularly large speed reductions on rail transit corridors (Lo and Hall 2006). This increase in delays indicates that higher quality service is particularly effective at reducing congestion.

Figure 24 Regional, Central City and CBD Mode Shares (Pisarski 2006)



Although transit is typically just 1-3% of total regional mode share, it represents a larger portion of urban commuting (typically 5-10%) and an even greater share (typically 10-50%) of peak-period travel to major activity centers such as central business districts (CBDs) and campuses.

Bhattacharjee and Goetz (2012) found that throughout recent years, Denver traffic volumes grew less on roads in light rail corridors than elsewhere: between 1992 and 2008, vehicle-miles traveled increased 41% outside the light rail zones but only 31% inside, despite rapid land development there, although, Ransom and Kelemen (2016) challenge this conclusion. Similarly, Kim, Park and Sang (2008) found that after the Hiawatha LRT line was completed, peak-period traffic volumes on that corridor decreased while regional traffic grew. Aftabuzzaman, Currie and Sarvi (2010) estimate that in Australian cities, high-quality public transit provides congestion cost reductions worth \$0.044 to \$1.51 per transit-vehicle kilometer, with higher values on the most congested corridors.

A major study by Jeihani, et al. (2013) evaluated the travel impacts of transit-oriented development (TOD) in the Washington, D.C. and Baltimore metropolitan regions. Out of 1,473 total transportation analysis zones in those regions, they classified 107 TOD's, occupied by approximately 11% of regional residents. Their detailed analysis indicates that all else being equal (accounting for various demographic and geographic factors), TOD residents drive about 20% fewer annual miles than residents of other areas, and rely significantly more on walking, cycling and public transport for both commute and non-commute trips. Since the vehicle travel reductions tend to be concentrated on major urban corridors, they provide proportionately

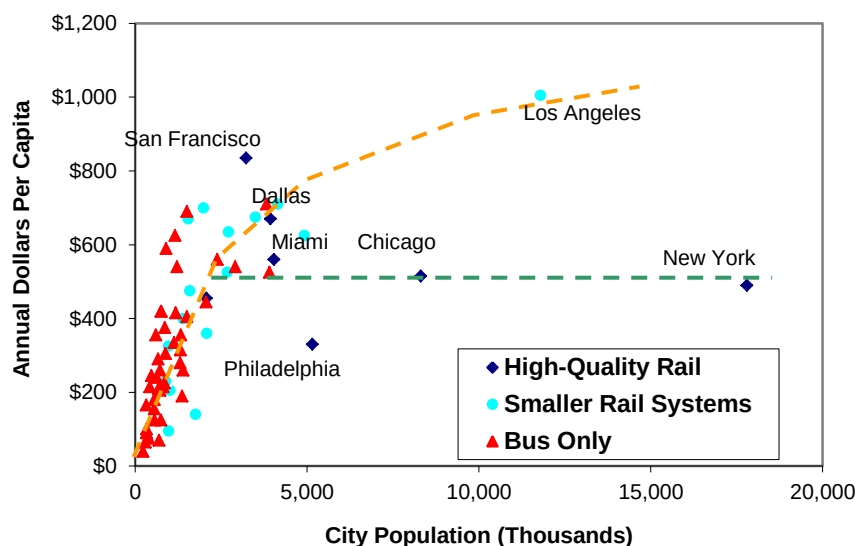
larger reductions in traffic congestion delays. Using a regional traffic model, they found that the TOD's 1.2% reductions in total regional vehicle travel reduce regional congestion delays by 2.8% and local delays by 20%, with similar air pollution emission reductions. During the PM peak period, TODs decreased 12,648 vehicle miles (0.41%), and 3,959 total hours of delay (4.0%).

Adler and van Ommeren (2016) analyzed the impacts of citywide public transit strikes in Rotterdam, The Netherlands. They found that a strike causes marginal weekday congestion increases on the highway ring road (0.017 minutes per kilometer), but substantially larger congestion on inner city roads (0.224 minutes per kilometer). These impacts are escalated during rush hour and diminished on weekends. Adler and Ommeren calculate that public transit's congestion relief benefit is equivalent to about half of its subsidy.

Ewing, Tian and Spain (2014) investigated the effects that Salt Lake City's University TRAX light-rail system has on vehicle traffic on parallel roadways. This rail system began operating in 2001 and expanded over the following years with new lines and stations. It currently carries about 53,000 average daily passengers. The study found significant declines in roadway traffic after the LRT line was completed, despite significant development in the area. The study estimates that the LRT line reduced daily vehicle traffic on the study corridor by about 50%, from 44,000 (if the line did not exist) to 22,300 (what currently actually occurs).

In a detailed economic analysis, Beaudoin and Lawell (2017) found that in U.S. cities, increases in public transit supply lead to a reduction in the demand for automobile travel and congestion, although efficient road pricing is required to maximize these benefits. They conclude that this benefit should be incorporated into transportation project economic evaluation. Studies by Garrett (2004) and Winston and Langer (2004) indicate that regional traffic congestion often declines as rail transit mileage expands. Cities with extensive grade-separated transit systems have lower per capita congestion costs than comparable size cities with lower quality transit services. For example, New York and Chicago have lower per capita congestion costs than Dallas and Los Angeles, as illustrated in Figure 25.

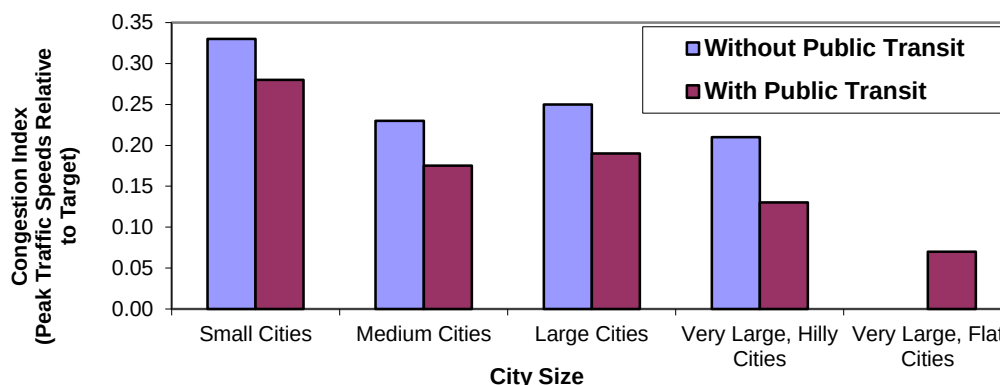
Figure 25 Congestion Costs (Litman 2004)



Traffic congestion costs tend to increase with city size (orange dashed line), except for cities with high-quality rail systems (green dashed line).

Similar patterns are found in developing countries. Figure 26 shows that Indian cities with rail transit have less intense roadway congestion.

Figure 26 Traffic Congestion in India (Wilbur Smith 2008)



Traffic congestion is lower in Indian cities with higher quality public transit.

The Texas Transportation Institute's (TTI) *Urban Mobility Reports* estimate the congestion reductions provided by public transit, based on the estimated increase in urban-peak traffic volumes that would occur if current transit trips shifted to automobile travel. Harford (2006) used data from the TTI reports to estimate the monetized value of transit congestion and pollution reductions as well as user consumer surplus gains; he estimated that these benefits provide a benefit–cost ratio of 1.34, with lower values in smaller urban areas and higher values in larger urban areas.

Some researchers claim that public transit fails to reduce traffic congestion, but their analyses do not reflect best practices. For example, Rubin and Mansour (2013) found a positive relationship between transit ridership and congestion, but they measured congestion *intensity* rather than *costs* (which ignores the congestion costs avoided by travelers who shift mode or have shorter trips), failed to account for confounding factors (city size and density, transit service quality, and employment rates), and aggregated all types of transit (Litman 2014b). Similarly, Duranton and Turner (2009) claim that transit fails to reduce congestion based on regression analysis of regional transit supply (buses and light rail carriages per 10,000 population) and average annual daily traffic (AADT) on regional highways. They fail to account for service quality (they do not differentiate between grade-separated and mixed traffic service), and since only a small portion of total traffic occurs under congested conditions, the AADT is a poor indicator of congestion. These studies do not prove that appropriate transit improvements on major urban corridors are ineffective at reducing congestion.

Walking, cycling and public transit improvements can also help reduce congestion costs indirectly by providing a catalyst for more compact development, which leverages additional vehicle travel reductions (Cortright 2010). Where this occurs, each additional transit passenger-mile typically reduces two to ten motor vehicle-miles (ICF 2010; Litman 2007). High-quality transit also complements decongestion pricing, it reduces the toll required to achieve a given reduction in traffic volumes and congestion delays (Parsons Brinckerhoff 2013; PSRC 2008).

Critics sometimes argue that because walking, cycling and public transit travel tends to be slower than automobile travel, travelers who shift from driving to space-efficient modes are worse off. However, travelers have diverse needs and preferences; travel decisions involve complex trade-offs between various benefits and costs. Travelers sometimes prefer slower modes for reasons such as they enjoy walking and cycling and appreciate the exercise benefits, or because they find public transit travel less stressful or more productive (they can rest or work) than driving on congested roads. If alternative mode improvements attract travelers out of automobiles, they must be directly better off, (increased consumer surplus) or they would not shift.

How congestion is evaluated significantly affects the estimated congestion reductions of transit improvement projects. For example, if evaluated using roadway level-of-service or the Travel Time Index, a general-traffic-lane to bus-lane conversion is only justified if a decline in general traffic delay occurs from the number of drivers who switch to transit methods. If evaluated based on per capita congestion costs, such a conversion is justified if, after completed, the bus-lane would carry at least as many peak-period passengers as a traffic lane (e.g. 800 on an arterial or 2,000 on a limited access highway). There is justification because bus passengers' time savings will exceed incremental automobile occupant delays. This discrepancy between the two methods occurs because the roadway level-of-service and the travel time index measure *vehicles*, while congestion cost indicators measure *people*, and so recognizes the additional savings and benefits that result if higher-occupant vehicles are given priority in traffic.

Transport Pricing Reforms

Transportation pricing reforms, such as those listed in Table 20, can reduce congestion by reducing vehicle traffic volumes and generating revenues for congestion reduction programs.

Table 20 Transport Pricing Reforms (Spears, Boarnet and Handy 2010; VTPI 2009)

Type	Description	Travel Impacts	Congestion Impacts
Decongestion pricing	Road tolls that are higher under congested conditions.	Reduces peak vehicle travel by shifting travel time, route, mode and destinations.	Provides large congestion reductions.
Flat tolls and vehicle travel fees	Tolls and mileage-based vehicle fees intended to generate revenue.	Shifts automobile travel to other modes and destinations. Reduces total vehicle travel.	Effects are dispersed. Provides modest congestion reductions.
Efficient parking pricing	Parking fees with higher rates and times and places with high parking demands, and variations such as parking <i>cash out</i> .	Shifts driving to other modes and destinations. Reduces total vehicle travel.	Because this is implemented most in dense urban areas, it tends to provide large congestion reductions.
Fuel tax increases	Increase fuel prices to generate revenue and internalize external costs.	Shifts driving to other modes and reduce total vehicle travel. Increases vehicle fuel economy.	Effects are dispersed. Provides modest congestion reductions.
Distance-based pricing	Prorate vehicle insurance premiums and registration fees by mileage.	Shifts automobile travel to other modes and destinations. Reduces total vehicle travel.	Impacts are potentially large but dispersed, so congestion reductions are modest.

This table summarizes major pricing reforms and their travel and congestion reduction impacts.

Decongestion pricing refers to road tolls with higher fees under congested conditions to reduce peak-period traffic volumes. This increases efficiency by allowing higher-value trips (urgent errands, commercial and high occupancy vehicle travel) to avoid congestion delays, and increases roadway operating efficiency (more vehicles per lane-hour). However, this type of pricing tends to have high implementation costs, raises privacy concerns, and only affects a minor portion of total vehicle travel. Other pricing strategies (flat road user fees, efficient parking pricing, higher fuel prices and distance-based pricing) affect a larger portion of total travel and therefore tend to be more effective at achieving other planning objectives such as reducing parking facility costs, crashes, energy consumption and pollution emissions.

Hybrid pricing strategies include *Value Pricing* (one highway lane is priced so motorists have an uncongested option) and *High Occupancy Toll* (HOT) lanes (lower-occupancy vehicles may use HOV lanes if they pay a toll).

Efficient parking pricing is an effective congestion reduction strategy. This can include expanding when and where parking is priced, changing monthly to daily or hourly fees, and offering off-peak discounts to encourage motorists to shift from peak to off-peak travel times (Primus 2018). Ostermeijer, et al. (2022) found that increasing in on-street parking prices for the city of Amsterdam caused substantial reductions in both parking and traffic congestion including a 2%–3% reduction in total vehicle traffic with particularly large declines during the evening peak.

Currently, road tolls are applied primarily to repay roadway construction costs. Where travel demand is sufficient tolls can finance roadway expansions, but vehicle travel is relatively price sensitive, particularly where there are good alternatives such as grade-separated public transit (Litman 2013; Williams-Derry 2011). For example, a \$1.00 to \$2.00 toll reduced traffic volumes over the I-65 bridge in Louisville by half, from 130,000 to 65,000 vehicles per day, generating far less revenue than needed to fully pay for the facility (Cortright 2021). As a result, many toll road projects have failed to achieve their traffic and revenue projections (Prozzi 2011).

Transportation pricing reforms, particularly road tolls, are often criticized as excessive and unfair. There are various ways to define what road user fees are appropriate and fair (Eby, Roskowski and Puentes 2020):

- *What motorists normally pay.* Since most roads are untolled, any road toll can be considered unfair by this indicator.
- *The price needed to reduce traffic to optimal levels.* This method usually justifies moderate to high fees, depending on demand. This method supports the use of a portion of revenues to improve alternative modes (such as public transit) since this can reduce the price needed to achieve a given reduction in traffic volumes. One major study found that the price elasticity of automobile commute trips is four times higher than average (-0.16 versus -0.04) on corridors with the best transit service (PSRC 2008), indicating that motorists would pay lower tolls to achieve a given congestion reduction target if public transit service is improved.
- *Cost recovery for roadway expansions.* This is often quite high since urban highway expansions are often quite costly.
- *The marginal external cost of vehicle travel.* This is often quite high under urban-peak conditions since motor vehicle travel tends to impose a variety of external costs (costs of building and maintaining roads and parking facilities, plus congestion, accident and pollution costs imposed on other people).
- *Impacts on lower-income people.* Transportation pricing is often considered regressive (poor people pay more relative to their incomes) since a given fee represents a larger portion of income to lower-income motorists. Contrary to this common association, overall equity impacts depend on how prices are structured, the quality of travel options and how revenues are used. Lower-income residents tend to drive less than average, particularly on congested urban highways. Because these lower income residents drive less and contribute a larger portion of bus users, the regressivity of congestion is reduced (Kuntzman 2018). Road tolls and parking fees are generally no more regressive than other funding options. For example, road tolls tend to be less regressive than financing highways with general taxes (Schweitzer and Taylor 2008) and may be progressive overall if they fund improvements to alternative modes frequently used by lower-income travelers.

This indicates the need to clearly defining the perspective used to evaluate pricing equity: a price structure that seems fair from one perspective may be considered unfair by another.

Improving Traveler Information

Congestion costs partly results from variability and uncertainty: travelers cannot predict how much time they will need to make a particular trip and so must add “buffer” time (TTI 2012). Traveler information such as predictions and real-time reports on roadway conditions provided by highway signs, radio congestion reports, and special commercial services such as TomTom (www.tomtom.com) and INRIX (www.inrix.com) can reduce these costs. Such services allow travelers to predict travel speeds and avoid congestion problems. The information acquired by these services is particularly valuable for commercial travelers (freight and service vehicles, and other types of business travel) due to their relatively high travel time costs. Such services can significantly reduce congestion costs.

Although traffic condition information is already provided through various public and private services, additional improvements are possible which would further reduce congestion costs to individuals and businesses. For example, the European Commission’s real-time traffic information services aim to provide road users with useful, accurate and up-to-date information in the following areas (EC 2013):

- The road network
- Traffic circulation plans
- Traffic regulations (such as speed limits and access restrictions),
- Recommended driving routes
- Real-time traffic data (such as estimated travel times)
- Information about congestion, accidents, road works and road closures
- Weather conditions,
- Other relevant safety-related information (such as the presence of animals or debris on a road).

Smart Growth Development Policies

Smart Growth is a general term for various policies that create more compact, multimodal communities that result in residents owning fewer vehicles, driving less and relying more on space-efficient modes. There is debate concerning how Smart Growth affects congestion. Some people assume that higher densities increase congestion (Melia, Parkhurst and Barton 2011), but Smart Growth includes other features that reduce it, as summarized in the table below.

Table 21 Smart Growth Congestion Impacts

Smart Growth Feature	Congestion Impacts
Increased development density	Increases vehicle trips within an area, but reduces trip distances and supports use of space-efficient modes
Increased development mix	Reduces trip distances and supports use of space-efficient modes
More connected road network	Reduces the amount of traffic concentrated on arterials. Reduces trip distances. Supports use of space-efficient modes.
Improved transport options	Reduces total vehicle trips.
Transport demand management	Reduces total vehicle trips, particularly under congested conditions.
Parking management	Can reduce vehicle trips and support more compact development

Smart Growth includes many features that can reduce traffic congestion.

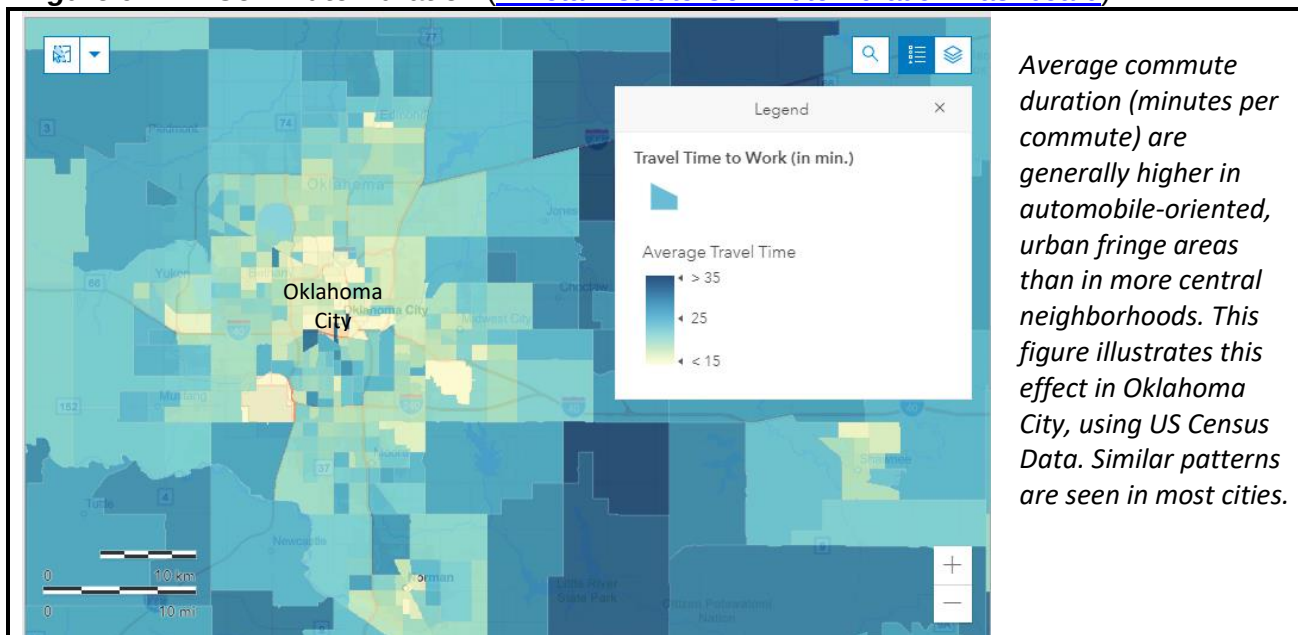
Ewing, Tain and Lyons (2018) found that compact development reduces but concentrates vehicle travel. These two effects roughly cancel each other out, so by itself, increasing development density typically has neutral impacts on per capita congestion costs. A major study in Phoenix, Arizona found less intense congestion and reduced per capita travel times in older neighborhoods with more compact and mixed development, more connected streets, better walking conditions and better public transit services in comparison to newer, lower-density, automobile-dependent suburbs (Kuzmyak 2012).

The benefits of compact development become evident when transportation performance is measured using accessibility-based indicators such as door-to-door travel speeds and total per capita transportation costs (Levinson and King 2020; Sundquist, McCahill and Dredske 2018). Although congestion intensity tends to be lower and traffic speeds tend to be higher in suburbs compared with central neighborhoods, suburban workers spend much more total time commuting to work than do residents of central areas since their faster traffic speeds are more than offset by longer travel distances, as indicates in Figure 9, which shows how average commute duration by all modes varies by geographic location. Suburban residents spend about twice as much time getting to work than in central areas, and these disparities are much larger for non-drivers who are forced to use slower modes and infrequent public transit services for those longer-distance trips.

This illustrates the difference between mobility-based and accessibility based evaluation: indicators of congestion intensity tend to justify urban highway expansions that may increase traffic speeds, at least temporarily, but if they stimulate more sprawl they often increase the total amount of time residents spend travelling, particularly for travellers who cannot, should not, or prefer not to drive for most trips. Conversely, despite lower traffic speeds, Smart Growth

policies improve overall accessibility and reduce the total distances that residents must travel to access services and activities.

Figure 9 Commute Duration ([Mineta Institute Commute Duration Dashboard](#))



TDM Programs

Various Transportation Demand Management (TDM) programs help reduce congestion, including employee transport management, transportation management associations and mobility management marketing (VTPI 2009). Such programs provide an institutional framework for implementing strategies such as rideshare matching and pricing reforms, and in various ways, encourage travelers to try efficient alternatives. Such programs tend to increase the effectiveness of other congestion reduction strategies.

TDM includes improved traveler information, including dynamic signs, maps, websites and mobile communications that allow travelers to anticipate, avoid and respond to delays. For example, a commuter who typically drives might adjust their schedule, route or mode to avoid congestion, or when stuck in unexpected congestion send a message to family or colleagues to warn of delays. Improving transit information can also make it easier and more desirable for drivers to switch to public transit.

Summary of Congestion Evaluation Strategies

Table 22 evaluates the impacts of five congestion reduction strategies and the degree to which they are considered in transport modeling and planning. Urban roadway expansions often provide only short-term congestion reductions and tend to increase other costs as well as only having few co-benefits. Conventional traffic models often exaggerate roadway expansion benefits, and conventional planning tends to favor this strategy. Other strategies tend to provide more long-term congestion reductions and more co-benefits but are often overlooked or undervalued in conventional transport modeling and planning.

Table 22 Congestion Reduction Strategies

	Roadway Expansion	Improve Alt. Modes	Pricing Reforms	Smart Growth	TDM Programs
Congestion impacts	Reduces short-run congestion, but this declines over time due to generated traffic.	Reduces but does not eliminate congestion.	Can significantly reduce congestion.	Increase local congestion intensity but reduces total regional congestion costs.	Can reduce congestion delays and the costs to users of those delays
Additional costs and benefits	High costs. By inducing additional vehicle travel and sprawl it tends to increase indirect costs. Minimal co-benefits. Small energy savings and emission reductions.	Moderate to high costs. Numerous co-benefits. Parking savings, safety and health, improved access for non-drivers, user savings, energy conservation, emission reductions, etc.	Low to high implementation costs. Costs users, creates revenue (economic transfers). Numerous co-benefits. Revenues, parking savings, traffic safety, energy conservation, emission reductions, improved public health, etc.	Low to high costs. Numerous co-benefits including infrastructure savings, safety and health, user savings, energy savings, emission reductions, improved non-drivers mobility, etc.	Generally low to moderate implementation costs. Numerous co-benefits.
Consideration in traffic modeling	Models often exaggerate benefits by underestimating generated traffic and induced travel	Models often underestimate the congestion reduction benefits of high quality space-efficient modes	Varies. Can generally evaluate decongestion pricing but are less accurate for other reforms such as parking pricing	Models often underestimate Smart Growth's ability to reduce vehicle travel and therefore congestion	Sometimes considered
Consideration in current planning	Commonly considered and funded	Sometimes considered, particularly in large cities	Sometimes considered but seldom implemented	Not generally considered a congestion reduction strategy	Sometimes considered, particularly in large cities

Different congestion reduction strategies have different types of impacts and benefits. Current traffic models and planning practices tend to undervalue many of these impacts.

Some strategies have synergistic effects; they are more effective if implemented together. For example, public transit improvements, efficient parking pricing and more compact development might individually only reduce vehicle travel 5%, but if implemented together, provide 30% reductions because their effects are complementary. Implementing these strategies as an integrated program maximizes their impacts and benefits.

What Does Modeling Indicate?

Most older traffic models are not very accurate at predicting long-term traffic congestion effects because they use fixed trip tables which assume the same number of trips will be made between locations regardless of the level of congestion between them (Volker, Lee, and Handy 2020). As a result, these models account for shifts in route and mode, and sometimes in time, but not in destination or trip frequency.

Newer models incorporate more factors and so are more accurate at predicting the travel and emission impacts of specific transportation and land use policies (Handy and Boarnet 2014). Johnston (2006) summarizes results from more than three dozen long-range modeling exercises performed in the U.S. and Europe using integrated transport, land use and economic models. These models indicate that the most effective way to reduce congestion is to implement integrated programs that include a combination of public transit improvements, pricing reforms and Smart Growth development policies. These studies indicate that a reasonable set of policies can reduce total vehicle travel by 10% to 20% over two decades, maintain or improve highway levels-of-service ratings (i.e., they reduce congestion intensity), expand economic activity, increase transport system equity (by distributing benefits broadly), and reduce adverse environmental impacts compared to the base case. Expanding road capacity, along with transit capacity, but without changing market incentives to encourage more efficient use of existing roads and parking, results in expensive transit systems with low ridership.

Puget Sound region modeling reached similar conclusions (WSDOT 2006). It found that neither highway widening nor transit investments by themselves are cost effective congestion reduction strategies. This conclusion was reached with the use of fixed trip tables that exaggerate the benefits of highway expansions and underestimates the value of transit system improvements. The most effective congestion reduction program includes both transit service improvements and road pricing; integrating these two gives travelers better options and incentives. Table 23 summarizes estimated congestion reduction benefits and project costs. Both have costs that exceed congestion reduction benefits, but transit improvements are more cost effective overall since they provide many additional benefits. These benefits include road and parking cost savings, consumer cost savings, crash reductions, improved mobility for non-drivers, energy conservation, emission reductions, and support for strategic land use.

Table 23 Congestion Reduction Economic Analysis (WSDOT 2004)

	Congestion Reduction Benefits		Direct Project Costs	
	Lower Estimate	Higher Estimate	Lower Estimate	Higher Estimate
Highway Expansion	\$1,500	\$2,200	\$2,500	\$3,700
Transit Improvements	\$480	\$730	\$1,200	\$1,500

This table indicates estimated highway and transit congestion reduction benefits and costs, in millions of annualized dollars. Neither approach provides congestion-reduction benefits that exceed costs, but transit provides many additional benefits.

Optimal Congestion Solutions

Comprehensive analysis, which considers various access factors, impacts, economic efficiency principles and social equity objectives, suggests that optimal congestion reduction involves the following steps:

1. Improve transport options (walking, cycling, public transit, ridesharing, carsharing and telecommuting) if there is demand. Target improvements on congested urban corridors, such as transit service improvements on congested roads, and commute trip reduction programs at major commercial centers. These widespread improvements reflect the principle of consumer sovereignty and can help reduce external costs such as traffic and parking congestion.
2. On congested roadways, favor space-efficient modes. For example, provide bus lanes on urban arterials if after all cost-effective transit service improvements and encouragement programs they would carry more than about 600 passengers per peak hour. If this threshold is met, a bus lane will carry more people than a general traffic lane. Similarly, provide High Occupant Vehicle (HOV) lanes on urban highways whenever they would carry more people than a general traffic lane. This increases efficiency.
3. If possible, apply decongestion pricing (tolls or fees that are higher during congested periods), priced to reduce traffic volumes to optimal levels (level-of-service C or D). Apply system-wide if possible, but if not, apply on the most congested highways and bridges, provided that it does not cause significant spillover problems.
4. Implement other transport pricing reforms to a politically feasible degree, including revenue generating tolls, efficient parking pricing, fuel price increases, and distance-based insurance and registration fees. These reforms are justified on various efficiency and social equity grounds. Increased revenues can be used to improve space-efficient modes (particularly public transit service improvements and fare reductions that reduce traffic congestion), help finance roadways, or reduce local taxes (they can be considered compensation for the impacts that urban roadways impose on adjacent communities).
5. Implement commute trip reduction and mobility management marketing programs, particularly in conjunction with improvements to space-efficient modes.
6. Only consider urban roadway expansions if, after all of the previous strategies are implemented, congestion problems are significant, and peak-period toll revenues would finance all associated costs. Peak period tolls test users' willingness-to-pay for the additional capacity. For example, if a roadway expansion would have \$5 million annualized costs, it should be implemented only if peak-period tolls on that road repay those costs. Off-peak tolls can be used to finance general roadway costs, such as maintenance and safety improvements, but not capacity expansion.

Some of these policies and investments, such as improvements to space-efficient modes and transportation demand management programs, might not be fully justified by congestion reduction benefits alone but are justified when all impacts are considered. External benefits that may not be usually considered include various savings and benefits, and social equity objectives since improving alternative modes ensures that non-drivers receive a share of transport improvement benefits, and user fees reduce subsidies that non-drivers contribute toward roads and parking facilities.

Examples

Many cities around the world are implementing innovative win-win solutions that reduce traffic congestion and help achieve other planning objectives (CAI-Asia 2007; Grant, et al. 2011; Nelson/Nygaard 2006; OECD/ECMT 2007; VTPI 2012). Examples are described below.

Pasadena, California commissioned a detailed study of potential traffic reduction strategies (Nelson\Nygaard 2006):

- Establish a target occupancy rate for on-street parking and develop a program to adjust meter prices to achieve that target.
- Develop Residential Parking Benefit Districts, with meter revenues dedicated to neighborhood benefits, including transportation demand management programs, transit pass subsidies, and local carsharing programs.
- Reduce or eliminate minimum parking requirements, and require parking unbundling (parking spaces are rented separately from building space, particularly for an apartment's second parking space) for new development in suitable locations.
- Encourage or require parking cash-out for new development and city employees (if parking is subsidized, employees can receive its cash equivalent if they do not drive).
- Establish funding for transit network improvement projects, and support efforts to establish a Bus Rapid Transit route.

Boulder, Colorado is a small city that has implemented a combination of transportation demand management strategies including improved walking, cycling and public transit services, campus transport management programs, and incentives to use these modes instead of driving when possible. Between 1995 and 2004 the downtown drive-alone rate declined almost 36%, from 56% to 36%, while the transit mode share has more than doubled from 15% to 34%.

Vancouver, Canada's transportation plan is based on these principles (Brown 2012):

- Accommodate travel demand growth using the existing road network, by improving alternatives to the car: transit, walking and cycling. Support regional measures to manage travel demand, such as carpooling, parking limits, bridge tolls and electronic road charges.
- Accommodate automobile travel, particularly in areas not well served by transit.
- Maintain good truck access without unreasonable impacts on local neighborhoods.
- Support traffic calming to reduce traffic speeds and prevent neighbourhood short-cutting.
- Support local retailing, personal, business and community services so that residents can find more of the services and jobs they need closer to home.

Based on these principles, the plan identified 70 major transport improvement initiatives, and set mode share targets for walking, transit, biking, and automobile travel. Vancouver's increased downtown housing supply through its "living-first strategy." Walking and cycling mode shares increased, now representing more than a third of all downtown trips, and automobile trips declined. From 1996 to 2011, regional population grew 18% and employment 16%, but total vehicle trips to and within the city declined about 5%.

More than 150 cities have implemented Bus Rapid Transit (BRT) systems which provide convenient, fast, comfortable and affordable urban bus services that attract discretionary travelers (*BRT Global Database*). For example, Bogotá, Columbia's TransMilenio system has 1,500 buses on dedicated bus lanes, plus 410 feeder buses. Seventy-five percent of Bogota residents rate the system as good or very good. The city has also developed an extensive pedestrian and bicycle path network, and many TransMilenio stations have large bicycle parking facilities.

Since decongestion pricing was introduced in central London in 2003, vehicle trips into the congestion pricing zone have declined by 17%, and congestion, measured as person-hours of delay per mile traveled, has fallen by 26%.

In 2002, Seoul, South Korea implemented various transport innovations including removal of a major downtown highway, development of a BRT system with more than 5,000 high-quality buses operating on 107 km of busways, and pedestrian and cycling improvements, plus a traffic control center which monitors traffic and parking problems on major arterials. This has greatly reduced congestion delay and accident risk.

The Marin County, California Safe Routes to School Program works to promote walking and biking to school. The program utilizes a multipronged approach to identify and create safe routes to schools while encouraging community-wide involvement. By its second year, the program was serving 4,665 students in 15 schools. Participating public schools reported increases in walking trips (64%), biking trips (114%), and carpooling trips (91%), and a 39% decrease in trips by private vehicles carrying only one student.

In 1993, Kunming, China established its *Public Transport Masterplan*, which gives priority to walking, cycling and public transport over private automobiles. The first bus lane opened in 1999, followed by a second in 2002, there was also the implementation of pedestrian and cycling improvements and Smart Growth policies that focus new development around railway stations. Resident satisfaction increased from 79% in 1999 to 96% in 2001.

In 1975, Singapore first implemented an Area Licensing Scheme (ALS) which required motorists to purchase a paper license before entering the central area. In 1998 the paper license was replaced by an automated Electronic Road Pricing (ERP) system which uses decongestion pricing to maintain optimal traffic speeds of 45 to 65 km/h on expressways and 20 to 30 km/h on arterial roads.

Many Asian cities have relatively few parking spaces, so motorists must often pay for using a parking space. In some cities, motorists must show that they have an off-street parking space before they are allowed to register a vehicle (Barter 2010). This strategy tends to reduce vehicle ownership and traffic and encourages the use of space-efficient modes.

Conclusions

Traffic congestion increases driver stress, travel time and vehicle operating expenses. There are many possible ways of measuring these costs. Which methods are used can significantly affect planning decisions.

Traffic congestion tends to maintain equilibrium: it increases to the point in which delays discourage additional peak-period vehicle trips. When confronted with congestion, some travellers choose their next-best alternative such as shifting when, how and where they travel. As a result, traffic congestion seldom becomes as severe as would be predicted if growth trends are extrapolated into the future, and urban roadway expansions seldom provide long-term congestion reductions because much of the additional capacity eventually fills with latent demand. The quality of travel and location alternatives affects incremental congestion costs, and long-term congestion reductions require changing this point of equilibrium.

Commonly-used evaluation methods tend to exaggerate congestion costs. Congestion is a moderate cost overall, smaller than vehicle, parking or crash costs, so a congestion reduction strategy would not be cost effective if it increases those costs, but is far more beneficial overall if reduces these costs or achieves other planning goals.

The traffic congestion costs that travellers bear and impose vary widely. Commercial, high-occupant vehicles, and urgent errands bear higher costs per vehicle-hour of delay. Travellers using space-efficient modes such as bicycling, bus and ridesharing impose less congestion per passenger-mile than automobile travel. Efficient road pricing and HOV lanes increase economic efficiency by favoring higher value trips and more efficient modes, an effect that is often overlooked in policy analysis.

Experts recommend the following practices for comprehensive congestion evaluation:

- Evaluate transport system performance based on overall *accessibility* (people's overall ability to reach desired services and activities) rather than just *mobility* (vehicle travel).
- Measure per capita congestion **costs** rather than **intensity**. Congestion intensity indicators fail to account for the amount residents drive during peak periods, and so undervalue strategies that improve transport options or reduce trip distances.
- Calculate congestion costs **imposed** by road users, rather than just the costs they **bear**. Use marginal congestion costing when calculating efficient road pricing and when comparing the costs of different modes, and therefore, the potential congestion cost savings of mode shifts.
- Measure delays to **all travelers**, not just to **motorists**. Account for the time savings to passengers from transit priority systems, and delays to walking and cycling caused by roadway expansions.
- Use **efficiency-optimizing** baseline speeds (LOS C), rather than **freeflow** speeds. Freeflow speeds reduce roadway capacity, making them expensive to maintain at all times. Efficiency-optimizing speeds maximize roadway capacity and fuel economy, and so are more realistic.
- Use **travel time values** that reflect users' actual willingness-to-pay for incremental speed gains. For value-priced lanes (lanes available for a fee), use consumer surplus analysis. For general travel time savings, willingness-to-pay is typically 20-40% of average wages for personal travel, and wages, benefits and equipment costs for commercial travel.

- Recognize **travel time cost variability**, and therefore, the efficiency gains provided by policies that favor higher value trips over lower-value trips. Accounting for this impact tends to increase the value of priced, freight and high-occupant vehicle priority strategies.
- Recognize that congestion tends to **maintain self-limiting equilibrium**: it increases to the point in which delays limit further peak-period vehicle travel. As a result, traffic volumes and congestion costs seldom increase as much as predicted by extrapolating past trends.
- Account for **generated and induced vehicle travel** when evaluating roadway capacity expansions. Induced travel tends to reduce predicted congestion reduction benefits, provides marginal consumer benefits and increases external costs.
- Use accurate **fuel efficiency** functions. Vehicle fuel efficiency generally peaks at about 50 miles per hour. Reducing severe congestion (LOS D-F) reduces fuel consumption and emissions, but reducing moderate congestion (LOS C) often increases these impacts, particularly over the long run if capacity expansion induces additional vehicle travel.
- Account for increased **crash costs** that result if congestion reductions lead to high traffic speeds.
- Account for **co-benefits** when evaluating potential congestion reduction strategies. In addition to reducing congestion, some strategies also reduce parking costs, provide consumer savings and affordability, improve non-drivers' accessibility, increase safety and health, reduce pollution emissions, and support strategic land use objectives.
- Recognize and account for **data collection biases**. For example, Inrix and TomTom indices oversample very congested roadways and so exaggerate congestion costs for average motorists.
- Evaluate impacts on **specific corridors**. Although space-efficient modes, such as public transit, may serve a small portion of total regional travel, their share is often large on major urban corridors, so auto-to-transit shifts can provide large congestion reductions.
- Discuss potential sources of **bias and variability** and apply sensitivity analysis.

Cost studies that ignore these principles tend to overestimate congestion costs. For example, the UMR's estimates reflect the higher bound of congestion costs. More realistic assumptions result in much lower estimates.

More comprehensive analysis tends to reduce the projected value of urban roadway expansions and recognize more benefits of improvements to space-efficient modes, efficient road pricing, transportation demand management programs, and Smart Growth development policies. Comprehensive analysis can help identify win-win solutions: congestion reduction strategies that help achieve other planning objectives.

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